I2SNet: A Lightweight Coarse-to-Fine Hierarchical Network for Hyperspectral Image Classification From Image to Subpixel

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***Abstract*—Hyperspectral image classification (HSIC) faces critical challenges, including restricted receptive fields in patch-level methods and coarse feature representations alongside subpixel blindness in conventional image-level networks. To address these bottlenecks, we propose I2SNet, a novel lightweight image-level end-to-end architecture driven by a coarse-to-fine hierarchical feature learning paradigm. The network systematically extracts spatial–spectral representations across patch, pixel, and subpixel scales. Specifically, a spatial–spectral linear transformer (SSLT) and a spatial–spectral deformable Mamba (SSDM) are unified to efficiently capture global patch interactions with linear complexity. Subsequently, multiscale spatial–spectral dynamic convolutions (MS^{3}DCs) extract fine-grained local contexts. Furthermore, a subpixel nonlinear mixing model (SNLMM) is introduced to explicitly address subpixel blindness by modeling the physical mixing mechanisms of diverse materials. Extensive experiments on four benchmark datasets demonstrate that I2SNet achieves strong classification accuracy while maintaining an efficient and compact computational profile compared to state-of-the-art approaches. To facilitate reproducibility, our code is publicly available at https://github.com/flyzzie/I2SNet**

***Index Terms*—Hyperspectral image classification (HSIC), spatial–spectral deformable Mamba (SSDM), spatial–spectral linear transformer (SSLT), subpixel nonlinear mixing model (SNLMM).**

I. Introduction

HYPERSPECTRAL image (HSI) is an important type of remote sensing image that captures hundreds of contiguous spectral bands, providing each pixel with a continuous and detailed spectral signature [1]. This rich combination of spatial and spectral information enables the precise identification of diverse land-cover materials for Earth observation applications [2], [3], [4], [5], [6]. Consequently, HSI classification (HSIC) has emerged as a fundamental task that assigns semantic labels to individual pixels to achieve comprehensive scene interpretation.

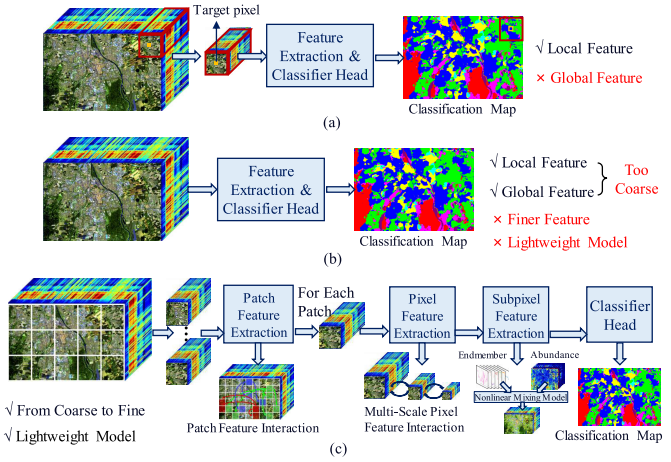


Fig. 1. Comparison of different spatial processing paradigms for HSI classification. (a) Patch-level methods. (b) Vanilla image-level methods. (c) Proposed I2SNet.

Driven by deep learning, HSIC has evolved significantly from handcrafted feature engineering to automated architectures. Early convolutional neural networks (CNNs) [7], [8], [9], [10], [11] demonstrated strong local feature extraction but struggled with global context due to constrained receptive fields. Subsequently, Transformer-based models [12], [13], [14], [15] addressed this by modeling long-range dependencies, albeit at the cost of quadratic computational complexity. Recently, state-space models (SSMs) such as Mamba [16], [17], [18], [19], [20], [21] have garnered increasing attention, offering a linear complexity alternative for global context modeling, with several advanced SSM approaches explicitly adopting separate spatial and spectral branches to process HSI data more effectively.

Despite continuous architectural innovations, existing HSIC methods are primarily divided into patch- and image-level frameworks based on their spatial processing paradigms. As illustrated in Fig. 1(a), patch-level [17], [22], [23], [24], [25] methods construct training datasets by employing an overlapping sliding window strategy to extract localized spatial–spectral cubes centered on individual target pixels

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as independent samples. While effective for local feature extraction, their inherently restricted receptive fields hinder the capture of global spatial dependencies. Conversely, as shown in Fig. 1(b), image-level methods [16], [26], [27] process the entire scene simultaneously to execute direct pixel-to-pixel mapping. However, vanilla image-level networks suffer from several bottlenecks. First, although they can capture both local and global contextual information, the resulting feature representations are often too coarse. The lack of multiscale feature processing in image-level networks leads to insufficiently fine-grained feature representations. Moreover, they exhibit severe subpixel blindness. Since mixed pixels are ubiquitous in real-world scenarios, relying solely on pixel-level contextual modeling while completely ignoring the underlying physical mixing mechanisms significantly degrades their discriminative capability in complex environments. Additionally, these vanilla image-level architectures lack a lightweight design, often leading to large parameter counts and high computational costs that hinder training and inference efficiency.

To explicitly address these bottlenecks, we propose a novel lightweight image-level end-to-end architecture named I2SNet. As presented in Fig. 1(c), this network is driven by a coarse-to-fine hierarchical feature learning paradigm that systematically extracts representations from raw HSI across patch, pixel, and subpixel scales. Different from existing methods that either rely on local patch-level processing or directly perform coarse image-level mapping, I2SNet preserves the global scene modeling ability of image-level networks while introducing a unified patch-pixel-subpixel representation hierarchy. Initially, the model performs patch feature extraction to establish broad spatial–spectral interactions, thereby compensating for the limited receptive field of conventional patch-based methods. These representations are subsequently refined through multiscale pixel feature extraction to capture fine-grained local contexts and alleviate the coarse representation problem of vanilla image-level networks. Crucially, the architecture further delves into the subpixel level by incorporating a nonlinear mixing model (NLMM) to explicitly extract physical endmembers and abundances for addressing mixed pixels. Therefore, the uniqueness of I2SNet lies in its progressive integration of global semantics, local details, and physical subpixel cues within a single lightweight image-level framework. Consequently, I2SNet supports accurate scene interpretation while maintaining a compact computational profile. The major contributions of this article are summarized as follows.

1) We propose I2SNet, a novel coarse-to-fine hierarchical network for HSIC. By integrating patch, pixel, and subpixel representations, it effectively overcomes the restricted receptive fields and coarse feature extraction issues of existing paradigms.
2) We design a robust spatial–spectral feature extraction pipeline. It unifies a spatial–spectral linear transformer (SSLT) and a spatial–spectral deformable Mamba (SSDM) to capture global patch interactions and employs multiscale spatial–spectral dynamic convolutions (MS^{3}DCs) to extract fine-grained pixel-level contexts.
3) We introduce a subpixel NLMM (SNLMM) block to address subpixel blindness. Guided by an aggregated abundance prior, this module models physical mixing mechanisms to generate discriminative subpixel features for accurate scene interpretation.
4) Extensive experiments on multiple benchmarks demonstrate that I2SNet achieves strong classification accuracy while maintaining a lightweight and computationally efficient architecture compared to state-of-the-art approaches.

The remainder of this article is organized as follows. Section II reviews the related work. Section III describes the architecture of the proposed I2SNet model. Section IV presents the experimental results and analysis on four benchmark hyperspectral datasets. Section V concludes this article and discusses future research directions.

II. Related Work

A. HSIC Methods

Early methods typically employed traditional machine learning algorithms such as maximum likelihood estimation [28], spectral angle mapping [29], random forests [30], and support vector machines [31] to classify spectral features. Since these approaches relied solely on single-pixel spectral signatures while neglecting surrounding spatial contexts, their performance was significantly limited. To address the presence of mixed pixels, several traditional methods approached HSIC from a subpixel perspective. For instance, Lu et al. [32] proposed a framework that integrates complementary information across subpixel, pixel, and superpixel levels through feature and decision fusion to enhance the discrimination of mixed spectral pixels. Furthermore, Xu et al. [33] introduced subpixel component analysis, which performs sparse image decomposition on abundance maps to extract subpixel attribute features.

With the advancement of deep learning, Han et al. [22] first introduced subpixel features into neural network frameworks by proposing a two-branch architecture for simultaneous unmixing and classification that delivers abundance information directly to the classification task. They subsequently developed SV^{2}Net [23] to further account for spectral variability and improve classification accuracy. Concurrently, Transformer architectures have been extensively explored for their global contextual modeling capabilities. Xu et al. [25] proposed a dual selection fusion network utilizing kernel selection and token selection blocks to adaptively learn optimal receptive fields and filter critical contexts for flexible multiscale spatial–spectral feature fusion. Li et al. [34] proposed a spectral coordinate Transformer that embeds coordinate blocks within the encoder and leverages rotary positional encoding to design novel masking patterns.

However, due to the quadratic computational complexity of standard Transformers, recent attention has shifted toward Mamba-based models, which offer efficient long-range dependency modeling with linear complexity. He et al. [17] improved the scanning sequence by designing a 3-D scanning order tailored for hyperspectral data. Zhao et al.

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[35] proposed a lightweight Mamba architecture based on a central pixel prior. Li et al. [16] introduced an image-level Mamba model to extract features from both spatial and spectral perspectives. Pan et al. [26] enhanced feature diversity by fusing local and global representations within a patch-level framework. Subsequently, Xu et al. [27] integrated a mixture of experts model to learn the correlation between various scanning directions and the directional distribution of land covers. Nevertheless, the fixed scanning orders employed by standard Mamba architectures lack the flexibility required to adaptively comprehend the irregular semantic distribution of complex land covers. To overcome this limitation, we propose deformable spectral and spatial scanning mechanisms that dynamically adapt to the intricate spatial distribution of various materials.

B. Linear Attention and Mamba

1) Linear Attention: Linear attention directly addresses computational challenges by fundamentally reducing operational complexity. Pioneering work [36] discarded the standard Softmax function in favor of a mapping function ϕ applied to the queries and keys, thereby reducing the computational complexity to $\mathcal{O}(N)$. However, this mathematical approximation often incurs a significant performance degradation. To address this, the Performer architecture achieves efficient attention computation through random feature mapping. Han et al. [37] proposed an agent attention mechanism that introduces a minimal set of agent tokens as intermediaries to sequentially aggregate and broadcast information. This approach significantly reduces computational costs while completely retaining global contextual modeling capabilities. Inspired by the Mamba architecture, MILA [38] integrates the macroscopic design principles of state-space models into linear attention. Meanwhile, architectures like EfficientViT [39] and Flatten Transformer [40] augment local feature extraction by strategically combining linear attention with convolutional operations.

In the domain of HSIC, Liang et al. [20] proposed a dual-branch Mamba-like linear attention network. By integrating dedicated spatial and spectral linear attention (Spectral-LA) modules, this network achieves efficient global dependency modeling while preserving linear computational complexity. Chen et al. [41] developed a spectral probe-guided linear attention mechanism that prioritizes discriminative bands to efficiently model long-range spectral dependencies and significantly lower computational overhead. Furthermore, Tan et al. [42] introduced an embedded linear vision Transformer featuring a linear-complexity structure to seamlessly capture global correlations among diverse multiscale semantic tokens. This architecture concurrently employs gated linear unit activation functions to strengthen the modeling of long-range relationships for accurate scene interpretation.

2) Mamba: Mamba [43] is built upon SSMs [44], [45], which model linear time-invariant systems mapping an input $\mathrm{x}(t)$ to an output $\mathrm{y}(t)$ through a hidden state $h(t)$. The continuous system, defined by parameters $(\mathbf{A}, \mathbf{B}, \mathbf{C})$, is discretized using a zero-order hold (ZOH) with timescale Δ to obtain the discrete system: $h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}\mathrm{x}_t, \mathrm{y}_t = \mathbf{C}h_t$, where $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \mathbf{B}$. While traditional SSMs use static parameters that limit contextual adaptability, the selective SSM (S6) makes $\mathbf{B}$, $\mathbf{C}$, and Δ dynamic functions of the input $\mathbf{x}$. This input-dependent parameterization allows the model to achieve selective perception and capture complex contextual information with linear-time complexity.

C. Nonlinear Mixing Models

The linear mixing model (LMM) [46] provides the classical physical basis for HSI unmixing. It assumes that each observed spectrum is a weighted linear combination of several endmembers, where the weights represent material abundances

$$\mathbf{Y} = \mathbf{EO} + \mathbf{Z} \tag{1}$$

where $\mathbf{E}$, $\mathbf{O}$, and $\mathbf{Z}$ denote endmembers, abundances, and noises, respectively. Xu et al. [47] used LMM-based material proportion (abundance) estimation to guide HSI super-resolution, demonstrating that abundance priors can benefit spatial–spectral reconstruction. Although LMM is simple and interpretable, it cannot sufficiently describe nonlinear scattering and complex material interactions in real scenes.

Therefore, NLMMs [48], [49], [50], [51] have been proposed to characterize complex physical interactions that conventional linear models fail to explain. In real-world scenarios, multiple scattering frequently occurs among ground objects with height differences, such as buildings and trees, before the reflected light reaches the sensor. The general form of an NLMM is represented as the sum of a linear mixture and a nonlinear fluctuation term

$$\mathbf{Y} = \mathbf{EO} + \Phi(\mathbf{EO}) + \mathbf{Z} \tag{2}$$

where Φ represents the nonlinear mapping function.

While early nonlinear models like bilinear and multilinear mixing models [49], [52] focus on low-order scattering, recent research has transitioned toward deep learning frameworks to approximate these complex mechanisms. For instance, EMLM-Net [53] utilizes a dual-stream autoencoder to explicitly model band-wise nonlinear intensities. For HSIC, S^2VNet [23] incorporates nonlinear AE-based unmixing and spectral variability modeling into a patch-level classification framework, showing that physically grounded subpixel information can provide discriminative guidance for classification.

From the perspective of HSIC, integrating such physical mixing priors into a data-driven framework allows the model to overcome subpixel blindness, providing highly discriminative abundance features for complex scene interpretation.

III. Proposed I2SNet Model

This section provides a detailed exposition of our proposed I2SNet model. First, we elucidate the overall framework of the I2SNet model and its workflow for HSIC. Subsequently, we delve into the intricate design of the key components within the I2SNet architecture.

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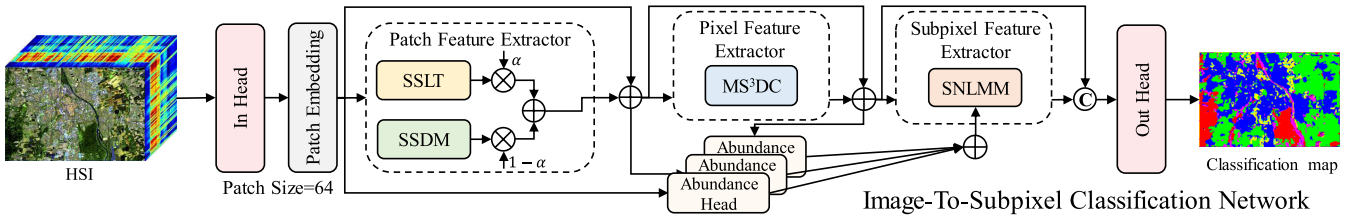


Fig. 2. Schematic of the proposed I2SNet model.

A. Framework of I2SNet Model

As illustrated in Fig. 2, the proposed model is an image-level end-to-end architecture that hierarchically extracts spatial–spectral representations from raw HSIs across patch, pixel, and subpixel scales. After initial processing via an in-head module comprising a point-wise convolution (PWC) and a 3×3 depthwise convolution (DWC), followed by patch embedding with a 64×64 patch size, the features enter a three-stage pipeline. First, the patch feature extractor captures global patch interactions using an SSLT and an SSDM in parallel, fusing them with learnable weights. Next, the pixel feature extractor extracts fine-grained multiscale contexts using MS^{3}DC. To address mixed pixels, multilevel features from preceding stages are processed through abundance heads to form a fused prior, which guides the SNLMM block. Finally, the SNLMM subpixel features and MS^{3}DC pixel features are concatenated and fed into the out head to produce the final classification map.

B. SSLT Block in Patch Feature Extractor

In this block, we optimize from both spatial and spectral perspectives by introducing spatial linear attention (Spatial-LA) and Spectral-LA, along with their integrated spatial–spectral fusion module (SSFM), which replaces the multihead self-attention mechanism in standard Transformer blocks.

In the upper half of Fig. 3, given an input feature map reshaped as a sequence $\mathbf{X} \in \mathbb{R}^{N \times C}$, where N is the total number of pixels and C is the spectral band dimension. It is important to note the spatial relationship $N = P \times P \times M$, where P denotes the spatial patch size and M represents the total number of patches. The SSLT block first projects $\mathbf{X}$ into shared queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ in $\mathbb{R}^{N \times B}$.

In the Spatial-LA branch, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are first spatially aggregated within each local patch through average pooling operations. This process generates representative patch-level tokens, yielding $\mathbf{Q}_{avg}, \mathbf{K}_{avg}, \mathbf{V}_{avg} \in \mathbb{R}^{M \times B}$. To capture the global correlations among these diverse patches, an initial attention map is constructed by computing the similarity between the averaged queries and keys, formulated as $\mathcal{M} = \text{Softmax}(\mathbf{Q}_{avg}\mathbf{K}_{avg}^{\top}/\sqrt{B})$. Subsequently, a sparsity constraint is enforced to explicitly model the most significant global similarities and suppress redundant noise. By applying a top-k masking operation on the initial attention map $\mathcal{M}$, we generate a sparse adjacency matrix $\mathcal{M}_{k_{spa}} \in \mathbb{R}^{M \times M}$. To further reduce computational overhead while preserving essential spatial interactions, the original queries $\mathbf{Q}$ and the averaged keys $\mathbf{K}_{avg}$ are projected into a lower dimensional space L, obtaining $\mathbf{Q}_{low} \in \mathbb{R}^{N \times L}$ and $\mathbf{K}_{low} \in \mathbb{R}^{M \times L}$. Finally, by incorporating the sparse global adjacency prior, the spatial attention output $\mathbf{O}_{spa}$ is calculated as

$$\mathbf{O}_{spa} = \text{Softmax}\left(\frac{\left(\mathbf{Q}_{low}\mathbf{K}_{low}^{\top}\right)\mathcal{M}_{k_{spa}}}{\sqrt{L}}\right)\mathbf{V}_{avg}. \tag{3}$$

Here, the multiplication between $(\mathbf{Q}_{low}\mathbf{K}_{low}^{\top})$ and $\mathcal{M}_{k_{spa}}$ denotes matrix multiplication. Specifically, $\mathbf{Q}_{low}\mathbf{K}_{low}^{\top} \in \mathbb{R}^{N \times M}$ is first multiplied by the sparse adjacency matrix $\mathcal{M}_{k_{spa}} \in \mathbb{R}^{M \times M}$, yielding an attention map in $\mathbb{R}^{N \times M}$. This map is then multiplied by $\mathbf{V}_{avg} \in \mathbb{R}^{M \times B}$ to obtain $\mathbf{O}_{spa} \in \mathbb{R}^{N \times B}$.

Parallel to the spatial branch, Spectral-LA is designed to capture global dependencies across the continuous spectral dimension. First, the queries $\mathbf{Q} \in \mathbb{R}^{N \times B}$ and keys $\mathbf{K} \in \mathbb{R}^{N \times B}$ are directly multiplied along the sequence dimension. This yields the spectral correlation matrix $\mathbf{S}_{spe} = \mathbf{Q}^{\top}\mathbf{K} \in \mathbb{R}^{B \times B}$. To emphasize the most informative spectral bands and filter out redundant information, a top-k selection is performed based on the row-wise sums of this correlation matrix, extracting the core k_{spe} spectral features, denoted as $\mathbf{E} \in \mathbb{R}^{k_{spe} \times B}$. By associating the spectral tokens through the proxy $\mathbf{E}$, the computational complexity is significantly reduced to linear with respect to N. The spectral attention output $\mathbf{O}_{spe} \in \mathbb{R}^{N \times B}$ is subsequently formulated as

$$\mathbf{O}_{spe} = \text{Softmax}\left(\left(\mathbf{Q}\mathbf{E}^{\top}\right)\text{Softmax}\left(\mathbf{E}\mathbf{K}^{\top}\mathbf{V}\right)\right) \tag{4}$$

where $\mathbf{Q}\mathbf{E}^{\top} \in \mathbb{R}^{N \times k_{spe}}$ and $\text{Softmax}(\mathbf{E}\mathbf{K}^{\top}\mathbf{V}) \in \mathbb{R}^{k_{spe} \times B}$.

To integrate the spatial and spectral representations, the SSFM illustrated in Fig. 3 employs a symmetric mutual cross-gating mechanism. The attention outputs $\mathbf{O}_{spa}$ and $\mathbf{O}_{spe}$ are initially reshaped into 3-D feature maps $\mathbf{X}_{spa}, \mathbf{X}_{spe} \in \mathbb{R}^{H \times W \times B}$. To dynamically generate the gating weights, the spatial branch applies PWC and the sigmoid activation function σ, whereas the spectral branch utilizes global average pooling (GAP) and σ. By defining a source feature branch index $i \in \{spa, spe\}$, these extraction operations are unified as follows:

$$\mathbf{W}_{spa}^{i} = \sigma\left(\text{PWC}\left(\mathbf{X}_i\right)\right), \quad \mathbf{W}_{spe}^{i} = \sigma\left(\text{GAP}\left(\mathbf{X}_i\right)\right) \tag{5}$$

where $\mathbf{W}_{spa}^{i} \in \mathbb{R}^{H \times W \times 1}$ and $\mathbf{W}_{spe}^{i} \in \mathbb{R}^{1 \times 1 \times B}$ denote the spatial and spectral gating matrices, respectively. These matrices are subsequently utilized to cross-calibrate the complementary features. Let $j \in \{spe, spa\}$ denote the complementary branch corresponding to branch i. The aggregated output $\mathbf{S}_i$ is then computed as

$$\mathbf{S}_i = \left(\mathbf{X}_j \otimes \mathbf{W}_i^{i}\right) + \left(\mathbf{X}_i \otimes \mathbf{W}_i^{j}\right) \tag{6}$$

where $\otimes$ represents element-wise multiplication.

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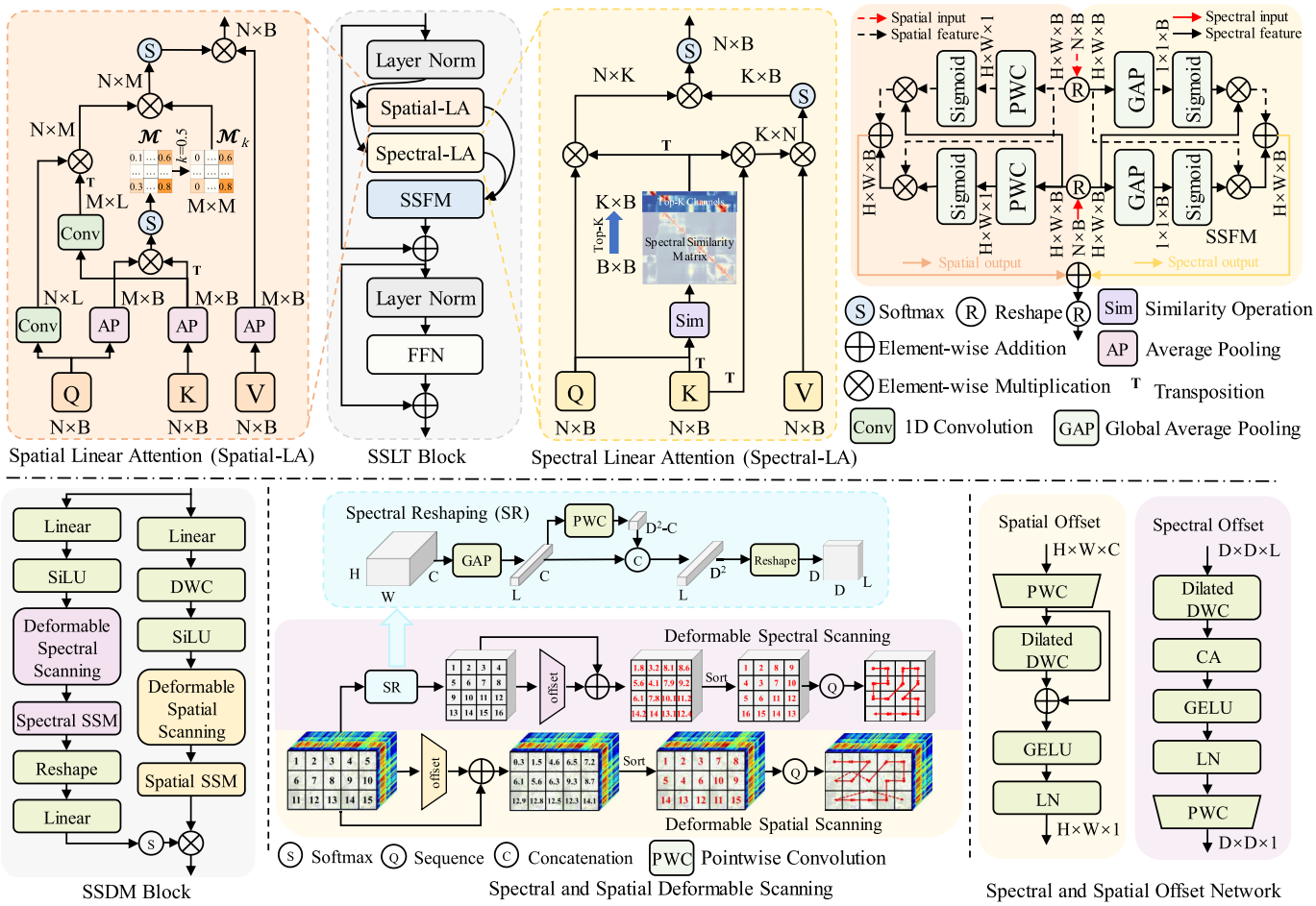


Fig. 3. Framework of SSLT and SSDM in patch feature extractor.

To conclude the fusion process, the aggregated features are summed and reshaped back to the original sequence dimension to form the final fused representation

$$\mathbf{O}_{fused} = \text{Reshape}(\mathbf{S}_{spa} + \mathbf{S}_{spe}) \in \mathbb{R}^{N \times B}. \tag{7}$$

Building upon the standard Transformer architecture, the complete SSLT Block encapsulates the aforementioned parallel attention branches and fusion mechanism alongside layer normalization (LN), residual connections, and a feed-forward network. Specifically, the intermediate representation $\mathbf{X}'$ is first derived by adding the input $\mathbf{X}$ to the fused attention outputs, formulated as $\mathbf{X}' = \mathbf{X} + \text{SSFM}(\text{Spatial-LA}(\text{LN}(\mathbf{X})), \text{Spectral-LA}(\text{LN}(\mathbf{X})))$. The final output $\mathbf{X}_{SSLT}$ of the SSLT block is then computed as

$$\mathbf{X}_{SSLT} = \mathbf{X}' + \text{FFN}\left(\text{LN}\left(\mathbf{X}'\right)\right). \tag{8}$$

This architectural design ensures the synergistic integration of spatial structural details and spectral correlations, thereby significantly enhancing the discriminative capability of the network while maintaining an ultralow parameter footprint.

To validate the theoretical efficiency of the proposed SSLT block, we provide a comparative analysis of its computational complexity against standard multihead self-attention. For an input sequence of length N and channel dimension B, the standard multihead self-attention mechanism requires the computation of a global attention map of size $N \times N$. This operation incurs a prohibitive quadratic complexity of $\mathcal{O}(N^2B)$, which scales poorly with high-resolution inputs. In contrast, the proposed Spatial-LA mechanism condenses the token sequence from N to a significantly smaller dimension M, where $M = N/P^2 \ll N$, through local patch aggregation. This structural reduction effectively limits the spatial attention complexity to $\mathcal{O}(M^2B + NMB)$. Concurrently, the Spectral-LA branch bypasses spatial pairwise interactions entirely by directly computing a $B \times B$ spectral correlation matrix. This yields a computational complexity of $\mathcal{O}(NB^2 + NBk)$, where k denotes the number of adaptively selected core spectral bands. Consequently, the overall computational complexity of the integrated branches scales linearly with respect to the spatial sequence length N. This linear scaling paradigm effectively circumvents the quadratic computational bottleneck inherent in standard Transformers, enabling the highly efficient processing of large-scale inputs.

C. SSDM Block in Patch Feature Extractor

To efficiently capture the continuous semantic features of land-cover and deep spectral dependencies with linear complexity, the SSDM block employs a dual-branch architecture. As illustrated in the lower part of Fig. 3, given an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, both branches leverage deformable

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scanning mechanisms to adaptively construct content-aware routing sequences for the SSM.

In the spatial branch, adopting the identical patch partitioning strategy as the SSLT block, the input features are fed into a deformable spatial scanning module to enable content-adaptive routing. Specifically, the features are initially processed by a PWC to compress the spectral dimension. This operation is followed by a dilated DWC featuring a residual connection that aggregates the PWC output with the DWC output. After applying GELU and LN, a terminal mapping layer generates the spatial offset tensor $\Delta\mathbf{S}_{spa} \in \mathbb{R}^{H\times W\times 1}$. This dynamic offset is subsequently added to a standard reference grid. A sorting algorithm is then applied to this deformed grid to determine the scanning indices, effectively reordering the spatial pixels into a 1-D sequence. This content-adaptive sequence is processed by the spatial SSM to yield the refined spatial representation $\mathbf{X}_{spa}$. The offset predictor provides a lightweight and context-aware mechanism for estimating scanning priorities. In the spatial branch, PWC reduces the spectral dimension before offset learning, and the dilated DWC enlarges the receptive field to capture broader neighborhood structures with limited cost. The residual connection preserves local responses, while nonlinear activation and normalization stabilize the predicted offsets. The spectral branch follows the same principle on the reshaped spectral pseudo-image. After being added to the reference grid, the predicted offsets define priority values for sorting the tokens into a 1-D scan sequence. Therefore, the deformation changes only the relative scanning order rather than performing coordinate resampling. Since each token is sorted once according to its priority, irregular deformations still yield a valid one-to-one sequence, and rare numerical ties are treated as equivalent priorities without token omission or duplication.

Parallel to the spatial pathway, the input $\mathbf{X}$ enters the deformable spectral scanning module, which initiates with a spectral reshaping operation. This step applies GAP to compress the spatial dimensions, yielding a 2-D feature matrix of size $L \times C$, where L denotes the number of classes in the dataset. To construct a 3-D grid suitable for SSM scanning, a PWC generates an auxiliary feature matrix of size $L\times(D^2-C)$. This auxiliary matrix is concatenated with the original matrix to form a comprehensive representation of size $L \times D^2$, which is subsequently reshaped into a $D \times D \times L$ pseudo-image.

To derive the spectral scanning trajectory, this pseudo-image is routed through a spectral offset network. The network sequentially applies a dilated DWC, a channel attention mechanism to emphasize salient bands, GELU, LN, and a PWC, ultimately predicting a spectral offset tensor $\Delta\mathbf{S}_{spe} \in \mathbb{R}^{D\times D\times 1}$. Analogous to the spatial branch, this offset modulates a reference grid to generate sorted scanning indices. The reordered spectral sequence is then processed by the spectral SSM. Following the SSM execution, the features undergo a reshaping operation and linear mapping, culminating in a Softmax activation to produce the final channel-wise attention weights $\mathbf{W}_{def}$.

To conclude the block operations, the representations from both branches are synergistically integrated. The spatial representation $\mathbf{X}_{spa}$ is recalibrated by the spectral attention weights $\mathbf{W}_{def}$ to yield the final output $\mathbf{X}_{SSDM}$

$$\mathbf{X}_{SSDM} = \mathbf{X}_{spa} \otimes \mathbf{W}_{def} \tag{9}$$

where $\otimes$ denotes element-wise multiplication.

This dual-branch design ensures that both dynamic spatial layouts and global spectral dependencies are modeled effectively, providing a highly discriminative feature representation guided by the proposed architectural framework.

D. MS^{3}DC in Pixel Feature Extractor

As shown in Fig. 4, to comprehensively capture fine-grained discriminative features at the pixel level, we propose the MS^{3}DC module. This module sequentially processes the input hyperspectral patch $\mathbf{X} \in \mathbb{R}^{P\times P\times C\times M}$ across both spectral and spatial dimensions. Focusing on a single patch $\mathbf{X}_{in} \in \mathbb{R}^{P\times P\times C}$, the module initially models spectral dependencies through a multiscale spectral dynamic convolution (MS-Spe-DC). The input $\mathbf{X}_{in}$ undergoes a channel shuffle operation followed by GAP to extract a global channel descriptor $\mathbf{X}_s$.

To capture spectral semantics at various granularities, this descriptor is processed by parallel PWCs at multiple scales and subsequently fused to recalibrate the spectral channels. The intermediate spatial–spectral feature $\mathbf{Y}_{spe} \in \mathbb{R}^{P\times P\times L}$ is computed as follows:

$$\mathbf{Y}_{spe} = \mathrm{PWC}_o\left(\mathbf{X}_{in} \circledast \mathrm{PWC}_f\left(\mathrm{Concat}_{j=1}^{J}\left(\mathrm{PWC}_j(\mathbf{X}_s)\right)\right)\right) \tag{10}$$

where PWC_j represents the jth parallel point-wise convolution generating intermediate vectors with progressively reduced channel dimensions. The operation $\mathrm{Concat}_{j=1}^{J}$ aggregates these multiscale vectors across J scales, which are then fused by PWC_f to form a comprehensive spectral attention vector. The symbol $\circledast$ denotes the standard convolution operation, and PWC_o serves to reduce the channel dimension from C to L.

The intermediate feature $\mathbf{Y}_{spe}$ is subsequently fed into the multiscale spatial dynamic convolution (MS-Spa-DC) branch to capture spatial patterns across hierarchical scales denoted by the index $i \in \{1, \ldots, I\}$. To maintain continuous cross-scale semantic routing, the robust input $\mathbf{F}_i$ for the kernel generator (KG) at each scale is recursively constructed as

$$\mathbf{F}_i = \begin{cases} \mathrm{DWC}_1(\mathbf{Y}_{spe}), & i = 1 \\ \mathrm{DWC}_i(\mathbf{F}_{i-1}^{\downarrow}) + (\mathbf{F}_{i-1}^{\downarrow} \circledast \mathcal{K}_{i-1}), & i > 1 \end{cases} \tag{11}$$

where $\mathbf{F}_{i-1}^{\downarrow}$ signifies the downsampled representation from the preceding scale $i-1$. Within the KG, the input feature $\mathbf{F}_i \in \mathbb{R}^{(P/(2^{i-1}))\times(P/(2^{i-1}))\times L}$ first undergoes a PWC and is bifurcated into parallel orthogonal pooling pathways. The horizontal X-Pooling and vertical Y-Pooling operations compress the spatial representations to dimensions of $1 \times (P/(2^{i-1})) \times L$ and $(P/(2^{i-1})) \times 1 \times L$, respectively. These pooled descriptors are subsequently mapped via linear projections to adjust their spatial sizes, yielding vectors of $1 \times k_i \times L$ and $k_i \times 1 \times L$. Finally, these two orthogonal vectors are multiplied to synthesize the scale-specific dynamic kernel $\mathcal{K}_i \in \mathbb{R}^{k_i\times k_i\times L}$.

Once the kernels for all scales are generated, they perform dynamic convolutions directly on the original intermediate feature $\mathbf{Y}_{spe}$ and are aggregated via element-wise addition. Concurrently, the multiscale cross-attention (MSCA)

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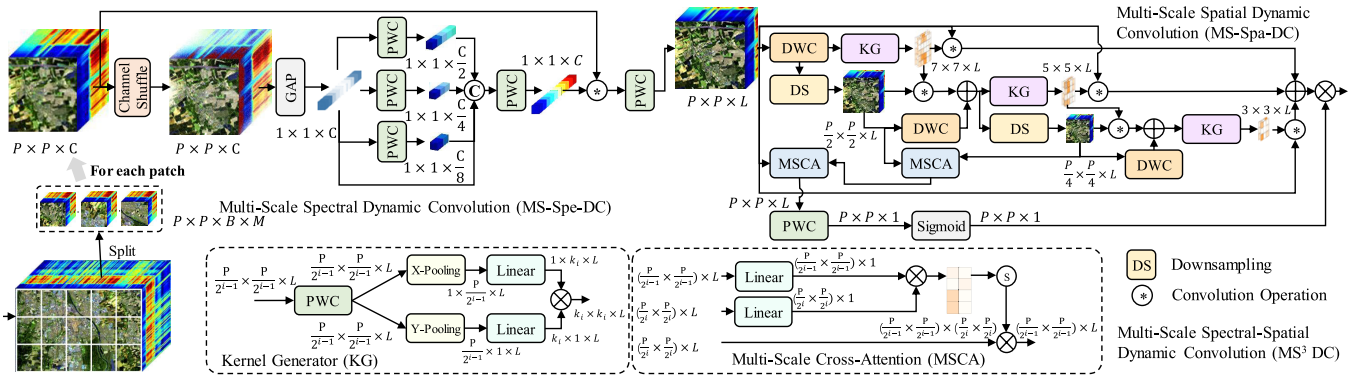


Fig. 4. Framework of MS^{3}DC in pixel feature extractor.

mechanism dynamically reweights features by computing attention matrices between adjacent spatial scales. Let $\mathbf{Z}_i \in \mathbb{R}^{((P/2^i)\times(P/2^i))\times L}$ and $\mathbf{Z}_{i-1} \in \mathbb{R}^{((P/(2^{i-1}))\times(P/(2^{i-1})))\times L}$ denote the features at the coarser scale i and the finer scale $i-1$, respectively. The cross-attention update is formulated as

$$\mathbf{Z}_{i-1} = \text{Softmax}\Big(\text{Linear}(\mathbf{Z}_{i-1})\text{Linear}(\mathbf{Z}_i)^\top\Big)\mathbf{Z}_i \tag{12}$$

where Linear$(\cdot)$ represents a linear projection that reduces the channel dimension from L to 1, effectively generating a cross-scale attention map of size $((P/(2^{i-1})) \times (P/(2^{i-1}))) \times ((P/2^i)\times(P/2^i))$. This interaction recursively cascades upward from the coarsest level to the primary resolution, yielding the final aggregated spatial feature $\mathbf{Z}_{msca} \in \mathbb{R}^{P\times P\times L}$. This interaction cascades upward to the primary resolution to yield an aggregated spatial feature $\mathbf{Z}_{msca}$. The final output of the MS^{3}DC module $\mathbf{X}_{out}$ is produced by applying an element-wise multiplication $\otimes$ between the aggregated multiscale convolution feature and the dense spatial attention map derived from $\mathbf{Z}_{msca}$

$$\mathbf{X}_{out} = \left(\sum_{i=1}^{I} \mathbf{Y}_{spe} \circledast \mathcal{K}_i\right) \otimes \sigma\big(\text{PWC}(\mathbf{Z}_{msca})\big) \tag{13}$$

where σ represents the sigmoid activation function. Different from ordinary multiscale convolution, MS^{3}DC does not simply stack several fixed kernels with predefined receptive fields. The MS-Spe-DC branch first recalibrates spectral responses through channel-shuffled multiscale descriptors, while the MS-Spa-DC branch generates scale-specific dynamic kernels from the current patch content. Therefore, the spectral and spatial receptive fields are adjusted according to material-dependent spatial–spectral variations. Compared with standard attention modules, MS^{3}DC preserves the local convolutional inductive bias and uses MSCA only to align and reweight cross-scale responses, avoiding dense token-to-token interaction over the whole patch. This design enhances fine-grained local structures with lightweight computation.

E. SNLMM in Subpixel Feature Extractor

To accurately decode the complex physical mixing processes at the subpixel level, as shown in Fig. 5, we introduce the SNLMM. As the final feature extraction stage within the architecture, SNLMM seamlessly couples multilevel abundance estimation, global–local endmember extraction, and a physically constrained NLMM.

The motivation for introducing SNLMM is that patch-level and pixel-level feature extractors mainly learn discriminative semantic representations, but they still regard each pixel as a single semantic unit. In real HSIs, however, one pixel often contains multiple materials due to limited spatial resolution and complex land-cover distributions. Relying only on global patch interactions or pixel-level local contexts may therefore ignore the physical composition inside mixed pixels. SNLMM is introduced to complement semantic feature learning with subpixel physical modeling. The abundance prior describes the fractional distribution of different land-cover components, the global–local endmember extraction estimates material-related spectral signatures, and the NLMM accounts for complex scattering and material interactions.

First, to establish a robust fractional distribution prior, the framework aggregates hierarchical features. Let $\mathbf{O}_1, \mathbf{O}_2, \mathbf{O}_3 \in \mathbb{R}^{H\times W\times L}$ denote the abundance logits derived from preceding stages. These intermediate predictions are fused via a softmax-normalized weighted combination to generate a comprehensive spatial abundance map $\mathbf{A}_{prior}$

$$\mathbf{A}_{prior} = \sum_{j=1}^{3} \omega_j \mathbf{O}_j, \quad \text{s.t.} \sum_{j=1}^{3} \omega_j = 1 \tag{14}$$

where ω_j are dynamic weights that adaptively balance coarse structural semantics and fine-grained subpixel details.

Here, $\mathbf{A}_{prior}$ should be regarded as soft abundance guidance rather than a hard prediction transferred from earlier stages. The logits $\mathbf{O}_1$–$\mathbf{O}_3$ retain uncertainty from different hierarchical levels and are fused by learnable weights before normalization. As a result, an incorrect response from a single preceding stage is not directly imposed on the subpixel unmixing process; it can be weakened by the other complementary logits and by the adaptive fusion weights.

Concurrently, SNLMM constructs a global endmember bank to capture the macroscopic spectral signatures. During initialization, pure pixels are extracted via training ground-truth masks to calculate the mean spectra, forming the global bank. The global signatures are then broadcasted to initialize the local spectral slots $\mathbf{Y}_0 \in \mathbb{R}^{L\times C}$ for each spatial patch.

To address intrinsic spectral variability, a patch-wise endmember extraction mechanism, inspired by slot attention [54], iteratively refines these slots. For a given input patch reshaped and normalized to $\mathbb{R}^{P^2\times B}$, parallel linear layers project it into keys $\mathbf{K} \in \mathbb{R}^{P^2\times B'}$ and values $\mathbf{V} \in \mathbb{R}^{P^2\times C}$. At iteration i, queries

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$\mathbf{Q} \in \mathbb{R}^{L\times B'}$ are projected from the previous state $\mathbf{Y}_{i-1}$. The intermediate spectral update $\mathbf{X}_{i-1} \in \mathbb{R}^{L\times C}$ is computed via cross-attention

$$\mathbf{X}_{i-1} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \tag{15}$$

To efficiently model the sequential dependencies of these continuous spectral updates, SNLMM employs a Gated S6 Block powered by SSMs. The block dynamically balances new spectral representations $\mathbf{X}_{i-1}$ with historical states $\mathbf{Y}_{i-1}$ through data-dependent discretized parameters ($\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C}$, and $\mathbf{D}$)

$$\mathbf{h}_i = \bar{\mathbf{A}}\mathbf{h}_{i-1} + \bar{\mathbf{B}}\mathbf{X}_{i-1}, \quad \mathbf{Y}_i = \alpha\mathbf{C}\mathbf{h}_i + (1-\alpha)\mathbf{D}\mathbf{Y}_{i-1} \tag{16}$$

where $\mathbf{h}_i$ is the hidden state. This recurrent process yields the final refined local endmembers $\mathbf{Y}_N$ after N iterations.

Finally, an NLMM closes the learning loop to impose physical constraints. The base linear reconstruction is computed as the matrix multiplication of the abundance prior $\mathbf{A}_{prior}$ and the final endmembers $\mathbf{Y}_N$. To model complex multiscattering and nonlinear interactive effects, this linear base is processed through a Feed-Forward Network and added back via a residual connection

$$\hat{\mathbf{F}} = (\mathbf{A}_{prior}\mathbf{Y}_N) + \text{FFN}(\mathbf{A}_{prior}\mathbf{Y}_N). \tag{17}$$

By minimizing the reconstruction loss between the generated HSI $\hat{\mathbf{F}} \in \mathbb{R}^{H\times W\times C}$ and the raw HSI, the SNLMM grounds the learned subpixel representations in physical spectral observations prior to final classification.

F. Loss Function

To effectively optimize the proposed I2SNet in an end-to-end manner, we formulate a joint loss function that simultaneously supervises the semantic classification and the physical spectral reconstruction. The total loss comprises a focal loss $\mathcal{L}_{cls}$ and a mean squared error (mse) loss $\mathcal{L}_{recon}$.

For the primary classification task, HSI datasets frequently suffer from severe class imbalance. To address this challenge and encourage the network to focus on hard-to-classify pixels, we employ the focal loss

$$\mathcal{L}_{cls} = -\frac{1}{\sum_{n=1}^{N} w_n}\sum_{n=1}^{N} w_n(1-p_n)^\gamma \log(p_n) \tag{18}$$

where N is the total number of labeled pixels and p_n denotes the predicted probability of the ground-truth class for the nth pixel. The focusing parameter γ is empirically set to 2. Additionally, the variable w_n represents the class-specific weight assigned to the nth pixel, which further mitigates the long-tail distribution issue.

For the subpixel unmixing process, we apply the mse loss to physically constrain the SNLMM module. By minimizing the discrepancy between the nonlinearly reconstructed image and the raw input, the network encourages the learned endmembers and abundance to remain grounded in physical reality

$$\mathcal{L}_{recon} = \frac{1}{N\times C}\sum_{n=1}^{N}\left\|\hat{\mathbf{F}}_n - \mathbf{X}_{raw,n}\right\|_2^2 \tag{19}$$

where $\hat{\mathbf{F}}_n$ and $\mathbf{X}_{raw,n}$ represent the reconstructed spectral vector (derived from the SNLMM output) and the raw input spectral vector for the nth pixel, respectively, with C being the number of spectral bands.

Finally, the joint loss function is formulated as the weighted sum of these two components

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda\mathcal{L}_{recon} \tag{20}$$

where the tradeoff hyperparameter λ is set to 0.01. This specific weight configuration ensures that the auxiliary physical reconstruction task regularizes the feature space effectively without dominating or interfering with the primary classification gradients.

IV. Experimental Results and Discussion

This section is organized as follows. Section IV-A details the four hyperspectral datasets employed to evaluate the proposed I2SNet model: Pavia University (PU), WHU-Hi-HanChuan (WHUHC), WHU-Hi-HongHu (WHUHH), and Augsburg. Section IV-B analyzes the impact of key hyperparameters, specifically the patch size and the hidden dimension after the input head, on overall model performance. Section IV-C conducts a comprehensive ablation study on the four proposed modules: SSLT, SSDM, MS^{3}DC, and SNLMM. It further evaluates different integration strategies between SSLT and SSDM to justify the parallel design of the patch feature extractor. Section IV-D conducts a comprehensive ablation study on the SSLT module within the patch feature extractor, alongside an analysis of the performance under various spatial and spectral top-k configurations. Section IV-E presents a thorough ablation analysis of the complementary SSDM module. Subsequently, Section IV-F describes the ablation study for the MS^{3}DC module, evaluating the classification performance across different spatial–spectral scales. Section IV-G reports on the ablation of the SNLMM module, investigating the effect of varying iteration numbers within the endmember extraction process. Finally, Section IV-H compares the classification performance and computational complexity of the proposed framework against several state-of-the-art network architectures.

First, we detail the parameter configurations and implementation settings adopted in our experiments. Initially, the spectral dimension of the input data is reduced to 64 within the In-head module, followed by a normalization process that scales all data values to the $[0, 1]$ interval. During the training phase, the model parameters are optimized using the Adam algorithm for a total of 100 epochs. The learning rate is empirically set to 0.0005. To stabilize the training process and prevent exploding gradients, gradient clipping is employed with a maximum norm of 1.0. To mitigate the influence of random variations and guarantee the statistical reliability of our findings, each experiment is executed independently ten times, with the average results reported.

To quantitatively assess the performance of the various classification methods, three standard evaluation metrics are employed: overall accuracy (OA), average accuracy (AA), and the kappa (κ) coefficient. Specifically, OA defines the total proportion of correctly classified samples across the

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entire dataset, AA calculates the mean classification accuracy across all individual categories, and the κ coefficient evaluates the statistical agreement between the predicted classification map and the ground-truth labels. Finally, all experiments are conducted on a hardware platform equipped with a 24-GB memory NVIDIA GeForce RTX 4090 GPU, utilizing a software environment configured with Python 3.7, PyTorch 2.2.0, and CUDA 11.7.

A. Hyperspectral Datasets

In our experiments, four benchmark hyperspectral datasets are used to evaluate the performance of our proposed I2SNet model. A detailed description is provided as follows.

1) Pavia University: The PU dataset, captured in 2001 by the ROSIS sensor over northern Italy, features a spatial size of 610×340 pixels and 103 spectral bands. It contains 42 776 labeled pixels categorized into nine distinct land-cover classes. In this experiment, 15 pixels from each class were selected as the training set, with the remaining pixels allocated for testing.

2) WHU-Hi-HanChuan: Introduced by Zhong et al. [55], the WHUHC dataset was acquired via a UAV-borne imaging sensor in June 2016 over a rural–urban fringe zone in Hubei, China. Offering rich spatial detail with centimeter-level resolution, the imagery features a spatial dimension of 1217×303 pixels and includes 274 spectral bands ranging from 400 to 1000 nm. In our experiments, 50 pixels per class were selected for the training set, while the remaining pixels were reserved for testing.

3) WHU-Hi-HongHu: Also introduced by Zhong et al. [55], the WHUHH dataset was acquired using a Headwall Nano-Hyperspec imaging sensor with a 17-mm focal length, mounted on a DJI Matrice 600 Pro UAV platform. The imagery features a spatial resolution of approximately 0.043 m and a spatial dimension of 940×475 pixels. It comprises 270 spectral bands spanning the 400–1000-nm wavelength range, encompassing 22 challenging land-cover categories. In our experiments, 50 pixels per class were randomly selected for the training set, while the remaining pixels were used for testing.

4) Augsburg: The Augsburg dataset was collected by the HySpex sensor over the city of Augsburg, Germany. The imagery features a spatial resolution of 30 m with a dimension of 332×485 pixels, encompassing 180 spectral bands that cover the wavelength range from 400 to 2500 nm. The dataset contains seven land-cover classes. In our experiments, we selected 146, 264, 21, 248, 52, 7, and 23 pixels for each respective class to form the training set, while the remaining pixels were allocated for testing.

B. Hyperparameters Analysis

In this section, we investigate the impact of the spatial patch size and the hidden dimension on both the classification performance and the computational complexity of the proposed model. As illustrated in Fig. 6(a)–(d), varying the spatial patch size from 16 to 64 exerts a negligible influence on the model's parameter count and computational cost in terms of FLOPs.

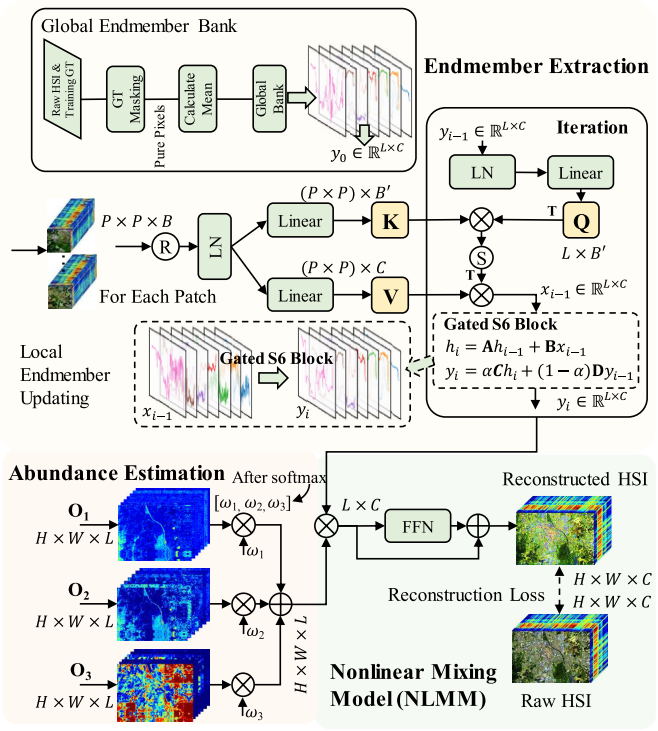


Fig. 5. Framework of SNLMM in subpixel feature extractor.

Consequently, the selection of this hyperparameter is exclusively driven by classification accuracy. For all subsequent experiments, we empirically set the spatial patch size to 64.

Conversely, the hidden dimension significantly dictates the tradeoff between model capacity and computational overhead, as shown in Fig. 6(e)–(h). Increasing the hidden dimension from 16 to 64 yields substantial improvements in OA across the evaluated datasets. However, further expanding the dimension to 128 substantially increases computational burden by approximately tripling both the parameters and FLOPs. Alongside this increased cost, the classification performance either saturates or noticeably degrades on most datasets, as evidenced by the accuracy dropping from 92.07% to 85.38% on the Augsburg dataset. Therefore, to strike an optimal balance between robust classification performance and computational efficiency, the hidden dimension is uniformly set to 64.

C. Ablation Study on Different Modules of I2SNet

As shown in Table I, to demonstrate the effectiveness of each proposed component, we conduct a comprehensive ablation study across the four hyperspectral datasets. The baseline network, consisting solely of the in and out heads, yields the lowest classification accuracy due to a lack of sufficient feature extraction mechanisms. We use **bold** to indicate the best performance and underline to denote the second-best results in the table.

First, integrating the SSLT and SSDM modules independently or simultaneously brings substantial accuracy improvements. The SSDM module specifically provides a clear performance gain by effectively modeling global coherent semantic correlations. Second, the inclusion of the MS^{3}DC module consistently enhances the classification results by

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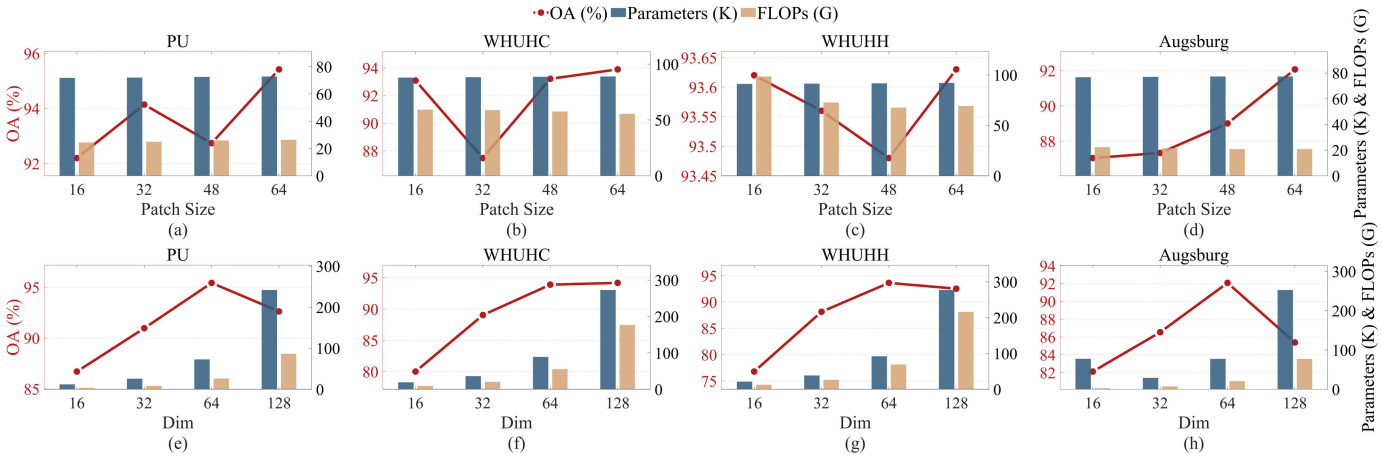


Fig. 6. OAs (%), parameters (K), and FLOPs (G) under different settings of hyperparameters. (a)–(d) Effects of spatial patch size on PU, WHUHC, WHUHH, and Augsburg, respectively. (e)–(h) Effects of hidden dimension on PU, WHUHC, WHUHH, and Augsburg, respectively.

TABLE I

ABLATION ON DIFFERENT MODULES OF I2SNET ACROSS FOUR DATASETS

No.	Patch		Pixel	Subpixel	PU			WHUHC			WHUHH			Augsburg		
	SSLT	SSDM	MS^{3}DC	SNLMM	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
1	×	×	×	×	76.44±0.98	81.53±0.44	69.81±1.07	67.42±0.97	62.89±0.78	62.81±1.05	64.38±0.86	69.18±0.43	58.95±0.75	81.80±0.51	48.97±1.48	72.28±0.79
2	✓	×	×	×	79.72±2.30	81.17±1.29	73.44±2.87	73.29±0.40	66.17±0.04	69.06±0.42	73.05±2.72	68.15±2.15	67.30±3.11	80.24±0.17	49.67±0.29	80.24±0.23
3	×	✓	×	×	90.16±0.51	91.48±0.14	87.17±0.59	88.41±1.31	87.15±1.24	86.50±1.51	87.75±1.41	86.29±1.45	84.74±1.75	80.49±0.24	48.07±2.57	71.49±0.57
4	×	×	✓	×	86.80±2.55	91.73±1.22	83.10±3.13	84.96±0.00	82.77±0.77	82.55±0.03	87.30±1.05	86.50±1.02	84.21±1.30	83.15±0.84	53.91±0.28	78.94±1.16
5	×	×	×	✓	78.92±0.52	82.06±1.90	72.05±0.84	68.22±1.69	61.19±1.76	61.44±1.85	63.85±0.34	67.32±2.29	58.12±0.07	87.69±0.42	59.18±0.66	82.11±0.63
6	✓	✓	×	×	90.89±0.28	94.18±0.49	88.09±0.42	91.93±0.57	90.61±0.60	90.58±0.65	91.08±0.39	91.23±1.08	88.88±0.51	87.30±0.65	62.66±0.16	81.61±0.92
7	✓	×	✓	×	88.73±0.05	92.80±0.46	85.34±0.01	87.41±0.02	87.39±0.47	85.36±0.02	91.01±0.79	89.71±0.33	88.76±0.97	86.99±1.82	63.38±3.02	81.20±2.66
8	✓	×	×	✓	78.59±3.50	81.50±1.19	72.13±4.26	77.01±1.01	69.91±1.50	71.71±1.16	75.11±0.86	67.05±3.22	69.44±1.22	87.09±0.77	62.67±0.04	81.44±1.10
9	×	✓	✓	×	92.01±1.12	94.12±0.59	89.59±1.41	91.08±0.56	89.09±0.84	89.59±0.61	90.70±0.19	90.83±0.22	88.39±0.25	85.11±0.38	55.54±1.02	78.51±0.71
10	×	✓	×	✓	89.01±0.83	90.87±0.87	85.70±1.05	89.09±1.21	87.33±1.54	87.28±1.40	88.11±1.19	88.02±1.18	85.23±1.42	80.47±0.04	49.53±3.93	71.47±0.29
11	×	×	✓	✓	88.95±3.36	93.18±0.78	83.05±4.03	85.06±0.26	83.53±0.35	82.66±0.29	87.93±0.62	87.93±1.54	85.00±0.81	86.12±0.47	59.75±4.25	79.92±0.90
12	✓	✓	✓	×	94.08±0.63	95.24±0.02	92.70±0.79	93.09±0.08	92.48±0.08	92.10±0.09	93.17±0.80	94.03±0.75	91.47±0.98	84.34±0.42	61.64±2.37	77.40±0.68
13	✓	✓	×	✓	92.73±0.82	95.23±0.53	88.92±1.07	91.72±0.68	90.67±0.53	90.35±0.79	90.11±1.73	90.84±1.29	87.72±2.10	87.08±0.37	62.80±1.38	81.30±0.59
14	✓	×	✓	✓	89.72±2.06	91.15±1.29	82.88±2.45	87.45±0.69	86.22±0.33	85.41±0.78	90.81±0.31	89.65±0.30	88.50±0.35	85.63±0.49	61.07±0.75	79.26±0.73
15	×	✓	✓	✓	92.08±0.06	93.17±0.19	88.37±0.10	89.41±1.12	88.51±0.26	87.67±1.28	90.45±0.45	90.74±1.34	88.08±0.59	89.07±1.60	62.30±1.23	84.54±2.21
16	✓	✓	✓	✓	**95.42±1.51**	**96.64±0.78**	**93.97±1.95**	**93.89±0.58**	**92.89±0.68**	**92.87±0.68**	**93.73±0.41**	**94.16±0.17**	**92.03±0.14**	**92.07±0.38**	**68.48±1.57**	**88.59±0.53**

aggregating comprehensive multiscale representations at the pixel level. Third, we observe a slight performance degradation in specific incomplete configurations involving the SNLMM module. This marginal drop in accuracy occurs because the incomplete preceding modules fail to extract sufficient features, thereby providing the SNLMM module with inadequate spatial representations for effective subpixel unmixing.

We further evaluate the synergistic effects of combinations involving exactly three modules to illustrate the specific contribution of each component. Omitting the SSLT module deprives the network of localized spatial structural awareness, while removing the SSDM module severely weakens the global coherent semantic correlation modeling, with both omissions resulting in noticeable accuracy drops. Similarly, bypassing the MS^{3}DC module limits the pixel-level multiscale aggregation capability and subsequently degrades the overall performance. These results show that different stages address complementary sources of error, where the patch-level modules strengthen semantic dependency modeling, MS^{3}DC recovers fine-grained pixel-level structures, and SNLMM introduces physical subpixel constraints. Although the configuration utilizing all feature extractors except the SNLMM module achieves highly competitive results, introducing the final physical unmixing stage remains crucial. Applying all four modules together constructs the complete I2SNet architecture and achieves the highest OAs on every dataset. This indicates that the hierarchical cues from patch-level extraction to subpixel-level unmixing are sequentially complementary rather than redundant, thereby maximizing the final classification performance.

TABLE II

COMPARISON OF DIFFERENT INTEGRATION STRATEGIES BETWEEN SSLT AND SSDM

	PU	WHUHC	WHUHH	Augsburg
	OA(%)	OA(%)	OA(%)	OA(%)
SSLT → SSDM	94.26±1.94	92.75±0.98	91.84±0.51	89.86±0.66
SSDM → SSLT	93.76±1.32	93.14±0.56	92.54±0.67	90.15±0.49
Parallel Concat	94.94±1.32	93.24±0.75	93.48±0.48	90.19±0.54
Parallel Add	95.01±1.01	93.55±0.76	93.46±0.55	90.97±0.25
Parallel Weighted Add	**95.42±1.51**	**93.89±0.58**	**93.73±0.41**	**92.07±0.38**

Since Table I has verified the individual effectiveness of SSLT-only and SSDM-only configurations, we further compare different integration strategies in Table II. SSLT and SSDM both enhance global spatial–spectral dependency modeling, but they capture different cues. SSLT is similarity-driven and explicitly models nonlocal correlations among spatial tokens and spectral channels, while SSDM is routing-driven and learns content-adaptive scan orders for sequential dependency modeling. Therefore, they are complementary rather than redundant. The serial variants, including SSLT → SSDM and SSDM → SSLT, improve the patch feature extractor but remain inferior to the parallel designs because the second

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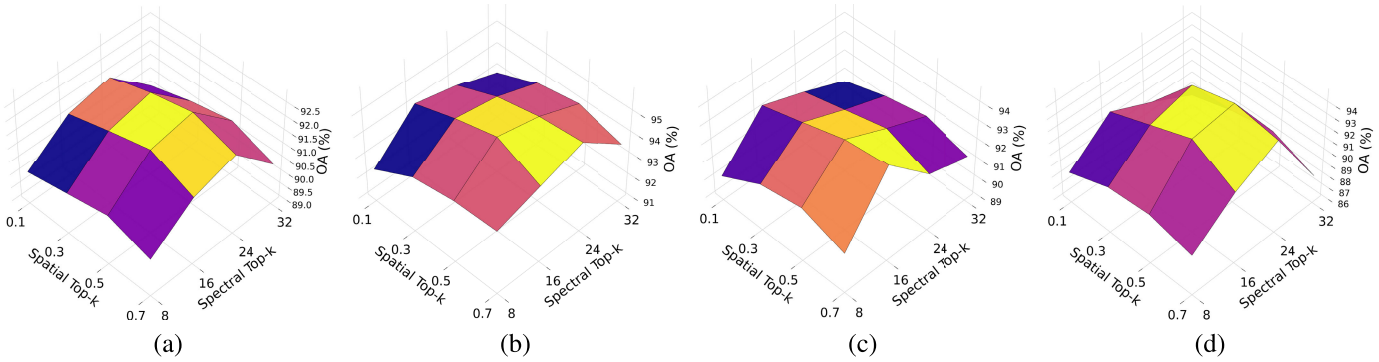


Fig. 7. OA (%) performance with respect to different spatial top-k and spectral top-k settings on (a) PU, (b) WHUHC, (c) WHUHH, and (d) Augsburg.

TABLE III
ABLATION STUDY ON SSLT OF PATCH FEATURE EXTRACTOR

	PU		WHUHC		WHUHH		Augsburg	
	OA(%)	GFLOPs	OA(%)	GFLOPs	OA(%)	GFLOPs	OA(%)	GFLOPs
w/o Spatial-LA	94.22±0.68	**22.19**	93.37±0.39	**47.84**	92.52±0.71	**60.08**	89.65±0.66	**18.77**
w/o Spectral-LA	94.13±1.82	23.65	93.82±0.08	48.67	92.16±1.37	61.09	88.62±1.80	19.17
w/o SSFM	94.39±0.98	26.35	93.24±0.37	55.39	92.43±0.80	69.14	90.07±0.22	22.39
w/ Spatial-VA	89.55±0.26	29.89	91.16±0.21	61.31	90.33±0.13	76.25	82.14±0.15	25.23
w/ Spectral-VA	90.26±0.16	33.12	91.04±0.25	62.15	90.44±0.22	77.26	83.16±0.31	25.64
w/ All-VA	90.34±0.34	36.25	92.24±0.38	68.02	91.24±0.27	84.31	84.24±0.37	28.46
w/ SSLT	**95.42±1.51**	26.37	**93.89±0.58**	55.42	**93.73±0.41**	69.17	**92.07±0.38**	22.41

TABLE IV
ABLATION STUDY ON SSDM OF PATCH FEATURE EXTRACTOR

	PU	WHUHC	WHUHH	Augsburg
	OA(%)	OA(%)	OA(%)	OA(%)
w/o Deformable Spatial SSM	89.00±0.03	90.43±1.73	90.60±0.41	90.41±0.28
w/o Deformable Spectral SSM	92.57±0.60	92.91±1.13	92.54±1.00	91.10±1.79
w/o Deformable Spatial Scanning	94.23±0.13	93.70±0.50	92.25±0.71	90.51±0.64
w/o Deformable Spectral Scanning	94.08±2.22	93.49±0.07	92.08±0.04	90.85±1.46
w/o Deformable Both Scanning	92.39±1.19	92.61±0.96	93.05±0.57	91.50±0.12
w/ SSDM	**95.42±1.51**	**93.89±0.58**	**93.73±0.41**	**92.07±0.38**

module can only process features already transformed by the first one. In contrast, the parallel design allows SSLT and SSDM to learn from the same patch-level input and preserve both similarity- and routing-aware information. Compared with simple concatenation and direct addition, the proposed parallel weighted addition achieves the best OAs on all four datasets, demonstrating that learnable parallel fusion provides the most effective patch-level representation for the subsequent pixel- and subpixel-level stages.

D. Ablation Study and Parameter Analysis on SSLT

As shown in Table III, to demonstrate the effectiveness of each component within the SSLT module, we conduct a comprehensive ablation study across the evaluated datasets. Removing either the Spatial-LA or the Spectral-LA consistently decreases the classification accuracy across all evaluated datasets. Replacing the SSFM with a simple addition operation similarly degrades the classification results. Furthermore, substituting the proposed linear attention mechanisms with vanilla attention results in a clear performance drop and a substantial increase in computational cost. Implementing vanilla attention across all components increases the computational burden on the PU dataset from 26.37 to 36.25 GFLOPs, while the OA decreases to 90.34%. This quantitative comparison demonstrates that our designed linear attention mechanisms are efficient and important for robust feature extraction. The improvement of SSLT comes from jointly modeling spatial nonlocal correlation and spectral band interaction. Spatial-LA links semantically related distant patches, Spectral-LA emphasizes informative spectral responses, and SSFM adaptively calibrates the two branches, thereby producing stronger global spatial–spectral features with controlled complexity.

Fig. 7 illustrates the qualitative impact of the spatial selection parameter k_{spa} and the spectral selection parameter k_{spe} on the network performance. The parameter k_{spa} dictates the proportion of the most relevant spatial patches retained within the sparse attention matrix. Meanwhile, the parameter k_{spe} controls the number of core spectral bands extracted based on the highest correlation scores. Increasing these top-k selection thresholds initially improves the classification results because the network incorporates a richer set of informative spatial contexts and discriminative spectral signatures. However, exceeding certain thresholds eventually introduces redundant noise and background interference into the feature representations. This overextraction increases the computational overhead without yielding further accuracy gains. To balance the model scale and the final classification performance, we empirically set the spectral selection parameter k_{spe} to 16 and the spatial selection proportion k_{spa} to 0.5 for all subsequent experiments.

E. Ablation Study on SSDM

Table IV presents the ablation study evaluating the individual components of the proposed patch feature extractor module. The complete removal of the spatial SSM branch severely degrades the classification accuracy. Specifically, eliminating this spatial component reduces the OA on the PU dataset from 95.42% to 89.00%. The spectral SSM branch also plays a substantial role in the feature extraction process since its omission decreases the OA on the WHUHC dataset from 93.89% to 92.91%. These quantitative outcomes confirm that both spatial and spectral branches are fundamentally necessary for robust HSIC. Beyond the basic parallel state-space architectures, substituting the proposed deformable scanning mechanism with a standard regular scanning strategy consistently limits model capacity. Reverting the spatial deformable scanning mechanism to a standard approach lowers the accuracy on the PU dataset to 94.23%. Similarly, excluding the spectral deformable scanning operation drops the OA on the

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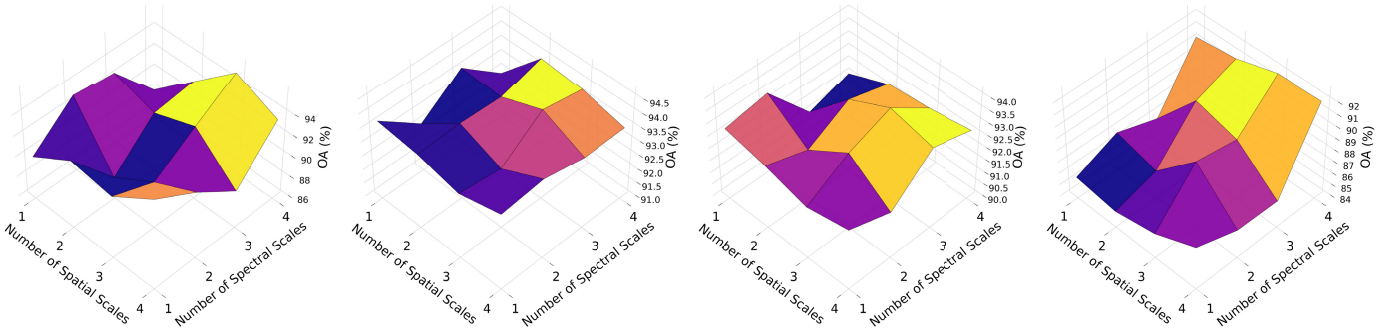


Fig. 8. OA (%) performance with respect to different spatial and spectral scale settings on (a) PU, (b) WHUHC, (c) WHUHH, and (d) Augsburg.

TABLE V
ABLATION STUDY AND REPLACEMENT COMPARISON ON MS^{3}DC OF PIXEL FEATURE EXTRACTOR

	PU			WHUHC	WHUHH	Augsburg
	OA(%)	Parameters (K)	FLOPs (G)	OA(%)	OA(%)	OA(%)
w/o MS-Spa-DC	93.18±0.84	65.48	0.370	92.75±0.17	92.87±0.79	91.64±1.31
w/o MS-Spe-DC	93.62±0.71	64.46	0.373	93.16±0.60	93.34±0.35	90.15±0.38
w/o channel shuffle	93.94±1.16	66.21	0.374	93.28±0.26	93.09±0.23	91.20±0.23
w/o MSCA	92.42±0.19	66.12	0.374	92.20±0.07	92.50±0.58	91.71±0.22
w/ MS^{3}DC	**95.42±1.51**	66.21	0.374	**93.89±0.58**	**93.73±0.41**	**92.07±0.38**
Fixed MS-Conv [56]	92.88±0.46	63.84	0.379	92.10±0.77	92.98±0.45	87.99±0.65
CBAM [57]	92.76±1.45	62.75	0.361	91.79±0.78	91.77±0.43	89.78±0.71

Augsburg dataset to 90.85%. Removing the deformable scanning technique from both branches simultaneously reduces the performance on the WHUHH dataset from 93.73% to 93.05%. The synergistic integration of both spatial and spectral SSMs equipped with deformable scanning mechanisms constitutes the complete module and achieves the optimal OA across all evaluated datasets.

The effectiveness of SSDM mainly comes from replacing fixed raster scanning with content-adaptive spatial and spectral routing. This allows Mamba to accumulate dependencies along more semantically coherent sequences, so long-range contextual information can be propagated more effectively.

F. Ablation Study and Parameter Analysis on MS^{3}DC

Table V details the quantitative ablation analysis of the MS^{3}DC in the pixel feature extractor. The initial channel shuffle operation effectively reorganizes the input spectral channels to promote cross-group information exchange before subsequent convolutions. Removing this operation significantly degrades the representational capacity of the network, dropping the OA on the PU dataset from 95.42% to 93.94%. Following the channel reorganization, the MS-Spe-DC generates adaptive weights across varying spectral receptive fields. Omitting this spectral component leads to a notable performance decline and reduces the accuracy on the Augsburg dataset by 1.92%. Similarly, the MS-Spa-DC extracts hierarchical contextual features through cascaded downsampling and DWC. Bypassing this spatial processing branch limits the structural awareness of the network and lowers the OA on the WHUHC dataset from 93.89% to 92.75%. Finally, the MSCA mechanism dynamically aligns and fuses the extracted hierarchical spatial features across different resolution levels. Discarding this interactive fusion weakens the interaction among multiscale representations and decreases the classification performance uniformly, evidenced by an accuracy drop to 92.20% on the WHUHC dataset and 92.42% on the PU dataset. The complete configuration integrates these operations to achieve the highest classification accuracy across all evaluated datasets. Although MS^{3}DC is slightly more complex in structure, we compare it with two lightweight replacement modules to verify the effectiveness of its dynamic spatial–spectral local modeling, including an Inception-style Fixed MS-Conv [56] and CBAM [57]. The parameters and FLOPs are measured on the PU dataset. Fixed MS-Conv uses fewer parameters than MS^{3}DC but has a slightly higher FLOP count, while CBAM has both fewer parameters and lower FLOPs. Nevertheless, both replacements achieve inferior overall classification performance on all four datasets. This gap is especially clear on the Augsburg dataset, where Fixed MS-Conv obtains an OA of 87.99%, which is much lower than the 92.07% achieved by MS^{3}DC. These results indicate that the improvement of MS^{3}DC does not come from simply increasing model size or using a generic lightweight module. Instead, adaptive spectral recalibration, content-dependent spatial dynamic kernels, and lightweight cross-scale alignment enable MS^{3}DC to preserve fine-grained local structures more effectively than fixed multiscale convolution or generic attention.

On the other hand, Fig. 8 visualizes the impact of varying the spatial and spectral scale levels within the MS^{3}DC module on the OA. Expanding these multiscale levels initially enhances the network performance by enabling the extraction of richer hierarchical spatial contexts and diverse spectral receptive fields. Exceeding certain scale limits eventually introduces redundant information and degrades the representational efficiency. Although the exact optimal scale combination fluctuates slightly depending on the specific characteristics of each dataset, configuring the spatial scale to 3 and the spectral scale to 4 consistently yields strong and stable results across the majority of evaluations. Consequently, we uniformly apply this scale configuration across all datasets to maintain a standardized and robust network architecture.

G. Ablation Study and Parameter Analysis on SNLMM

Table VI details the quantitative ablation analysis of the SNLMM. The endmember extraction process inherently relies on both macroscopic spectral stability and microscopic spatial variability. Removing the global endmember bank deprives

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TABLE VI
ABLATION STUDY ON SNLMM OF SUBPIXEL FEATURE EXTRACTOR

	PU OA(%)	WHUHC OA(%)	WHUHH OA(%)	Augsburg OA(%)
w/o Global Endmember Bank	94.82±1.12	93.65±0.38	93.48±0.52	90.25±0.35
w/o Local Updating	94.95±1.25	93.72±0.42	93.51±0.48	90.76±0.39
w/o $\mathbf{O}_1$	95.12±1.32	93.78±0.48	93.55±0.45	91.93±0.37
w/o $\mathbf{O}_2$	95.05±1.28	93.75±0.45	93.52±0.47	91.41±0.41
w/o $\mathbf{O}_3$	95.18±1.35	93.81±0.51	93.58±0.43	91.52±0.38
Linear Mixing	94.56±0.95	93.42±0.32	93.35±0.62	90.84±0.36
w/ SNLMM	**95.42±1.51**	**93.89±0.58**	**93.73±0.41**	**92.07±0.38**

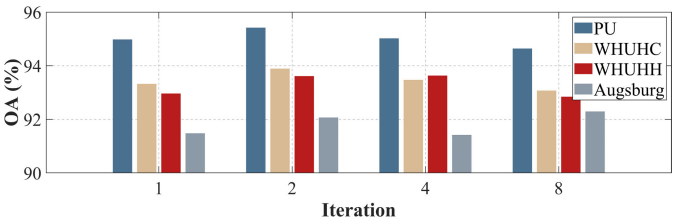


Fig. 9. OAs (%) on the PU, WHUHC, WHUHH, and Augsburg datasets with respect to a different number of iterations.

the network of foundational spectral signatures and causes a noticeable accuracy drop across all datasets. Similarly, discarding the local updating mechanism restricts the model from effectively addressing inherent spatial–spectral variations, thereby degrading the overall performance.

The framework establishes a robust fractional distribution prior by fusing multilevel abundance estimations. Omitting any intermediate prediction from the preceding stages disrupts the dynamic balance between coarse structural information and fine-grained subpixel details, which leads to suboptimal spatial abundance priors and classification degradation. The small accuracy degradation observed when removing a single $\mathbf{O}_j$ does not mean that the corresponding prediction is unnecessary. Instead, it indicates that the fused abundance prior is robust to noise or uncertainty from any individual preceding stage. Since $\mathbf{A}_{prior}$ is formed from multilevel soft logits rather than hard patch labels, possible early patch-level errors are not directly propagated as fixed subpixel labels. The improvement of SNLMM comes from converting high-level semantic cues into physically constrained subpixel representations, where the abundance prior, endmember extraction, and nonlinear reconstruction jointly help distinguish mixed pixels that are difficult to separate using discriminative features alone. Furthermore, replacing the physically constrained NLMM with a standard linear mixing strategy severely impairs the network. The OA on the Augsburg dataset drops from 92.07% to 90.84% when the nonlinear component is removed. This confirms that the nonlinear reconstruction constraint reduces the influence of unreliable early responses and helps the final subpixel unmixing representation better adhere to the raw spectral signatures.

Fig. 9 illustrates the influence of the slot attention iteration count on the final classification performance. A single iteration fails to sufficiently aggregate the local raw spectra and refine the patch-specific endmembers. Conversely, excessive iterations introduce oversmoothing into the continuous spectral updates and degrade the representational capacity of the network. The architecture consistently achieves the best OA across the evaluated datasets when the iteration count is set to two. Consequently, we configure the physical-anchored slot attention mechanism with two iterations for all subsequent experiments to ensure robust subpixel unmixing and maintain strong classification accuracy.

H. Classification Comparison With State-of-the-Art Models

Tables VII–X summarize the quantitative evaluation results of the proposed I2SNet against state-of-the-art patch- and image-level methods in terms of OA, AA, κ, and per-class accuracies. The proposed model achieves the highest OA, AA, and κ scores across all four benchmark datasets. Specifically, I2SNet yields OA improvements of 4.54%, 3.83%, 2.77%, and 1.40% over the respective second-best methods on the PU, WHUHC, WHUHH, and Augsburg datasets. In terms of per-class accuracy, I2SNet obtains the best or second-best results in most land-cover categories, although several minority or highly mixed categories remain challenging. To further substantiate the effectiveness of each model, we present a qualitative comparison of the generated classification maps in Figs. 10–13. In contrast to the baseline approaches, the maps generated by I2SNet display less salt-and-pepper noise along with smoother and more accurate regional transitions, showing good alignment with the ground truth. Particularly in the white-boxed area, our model achieves more refined discrimination among land covers, even under complex conditions with multiple clustered categories.

Furthermore, we evaluate the classification robustness of the proposed framework under limited training sample conditions. As illustrated in Fig. 14, decreasing the number of training samples inevitably degrades the OA across all compared models. However, I2SNet consistently maintains a clear performance margin over the competitors across different data sparsity levels. Even when only five training samples per class are used, the proposed method achieves an OA of 83.37% on the PU dataset and 80.27% on the WHUHC dataset. This stability indicates that the hierarchical spatial and spectral feature extraction combined with the physical unmixing mechanism improves the generalization capability and data efficiency of the network.

Beyond classification accuracy, Table XI demonstrates the computational efficiency of I2SNet. To ensure a fair comparison, both the computational cost and parameter count are evaluated on the PU dataset given an input size of $103 \times 128 \times 128$. Compared to all baselines, I2SNet achieves the lowest computational cost of 0.374 GFLOPs and a compact size of 66.21k parameters. Furthermore, it delivers the fastest execution speeds, requiring 51.32 s for training and 0.21 s for testing inference. This confirms that integrating linear attention with lightweight state-space models effectively mitigates the heavy overhead typical of standard Transformers. Consequently, I2SNet achieves a favorable balance between robust performance and computational economy, making it suitable for resource-constrained remote sensing applications.

I. Visualization of Internal Modules

This section visualizes the internal mechanisms of I2SNet, focusing on the feature enhancement of SSLT, the

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TABLE VII

OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR THE PU DATASET

	Patch-level						Image-level			
	DSNet [22]	SV^{2}Net [23]	DSFormer [27]	CSCANet [24]	3DSS-Mamba [18]	SliMamba [35]	MambaHSI [16]	MambaLG [26]	MambaMoE [27]	Proposed
Asphalt	88.86±3.32	83.38±7.70	77.78±2.37	78.85±2.50	54.98±26.82	76.73±10.21	88.32±3.80	83.35±3.30	88.05±1.20	**92.69±3.02**
Meadows	87.61±2.70	93.88±2.62	**96.59±1.61**	87.99±5.63	59.01±28.82	90.85±5.55	94.10±1.25	89.80±1.38	89.97±1.16	95.62±3.25
Gravel	76.51±7.35	74.39±13.51	87.37±2.08	85.24±5.91	59.00±15.20	87.38±7.18	85.11±4.88	79.77±5.06	**94.79±2.32**	94.64±1.88
Trees	87.75±3.85	91.19±3.45	85.29±4.25	91.62±2.41	62.43±14.44	83.62±6.62	90.11±1.68	94.19±0.33	95.16±1.87	**97.76±0.80**
Metal sheets	97.55±2.74	97.71±2.28	99.47±0.31	**100.00±0.00**	99.40±0.66	95.55±3.01	99.73±0.25	**100.00±0.00**	99.64±0.43	99.86±0.12
Bare soil	61.83±6.31	70.78±5.96	83.75±2.44	85.69±5.88	34.26±22.95	69.43±16.33	55.07±0.31	84.35±2.70	86.01±5.96	**91.99±5.39**
Bitumen	88.18±3.11	93.23±3.53	**99.62±0.25**	99.51±0.37	75.54±30.81	99.07±1.08	86.98±5.77	94.26±1.25	96.82±3.49	98.65±1.60
Bricks	67.70±3.89	81.70±8.88	67.81±2.18	85.64±12.09	58.30±20.34	66.48±13.26	90.61±4.96	92.39±0.05	93.95±2.03	**98.65±0.69**
Shadows	99.68±0.29	**99.91±0.08**	94.99±2.74	99.89±0.07	81.12±12.00	98.32±1.41	99.76±0.18	99.89±0.00	99.42±0.67	99.89±0.17
OA (%)	83.12±1.26	87.58±2.09	88.58±1.67	87.21±2.55	57.91±13.17	83.92±3.38	87.67±0.57	88.88±0.46	90.88±0.73	**95.42±1.51**
AA (%)	83.96±0.59	87.35±1.14	88.08±1.50	90.49±0.79	64.89±8.04	85.27±3.04	87.75±0.99	90.89±0.98	93.76±0.78	**96.64±0.78**
$\kappa \times 100$	77.73±1.66	83.49±2.68	84.94±2.15	83.42±3.09	49.26±13.31	78.89±4.31	83.46±0.75	85.46±0.66	88.07±0.96	**93.97±1.95**

TABLE VIII

OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR THE WHUHC DATASET

	Patch-level						Image-level			
	DSNet [22]	SV^{2}Net [23]	DSFormer [27]	CSCANet [24]	3DSS-Mamba [18]	SliMamba [35]	MambaHSI [16]	MambaLG [26]	MambaMoE [27]	Proposed
Strawberry	88.01±2.08	88.61±2.17	88.42±0.41	86.56±1.88	80.56±3.89	83.24±6.22	80.20±5.06	88.17±2.87	90.02±10.25	**92.64±1.18**
Cowpea	77.16±2.97	83.21±1.65	82.16±1.61	83.12±0.61	74.73±7.29	81.94±3.06	56.77±5.42	50.94±9.68	81.80±7.51	**86.40±3.34**
Soybean	84.57±2.88	90.84±2.61	81.94±1.95	85.29±2.56	89.18±5.46	87.22±3.56	84.82±3.15	91.97±3.54	94.53±9.89	**97.97±1.27**
Sorghum	95.04±2.38	97.20±0.41	**97.99±0.35**	97.98±0.23	95.98±0.56	96.06±2.51	95.87±1.43	96.21±1.99	96.07±2.48	96.00±1.22
Water spinach	97.29±2.13	98.54±0.82	94.64±4.12	96.84±1.62	97.14±0.62	97.33±2.04	99.58±0.42	**99.98±0.04**	99.93±0.16	99.97±0.07
Watermelon	60.86±7.30	70.78±8.74	85.19±3.55	85.90±2.51	76.19±5.00	85.07±4.30	51.89±7.01	48.81±6.80	82.46±23.59	**92.00±3.59**
Greens	88.58±3.19	90.96±2.12	93.02±4.52	91.96±1.32	92.99±3.16	94.30±1.28	90.38±10.96	89.88±3.05	83.18±14.90	**94.53±4.75**
Trees	67.32±2.39	76.99±5.09	80.38±2.45	79.51±4.04	70.20±10.08	74.40±2.50	58.54±8.50	53.98±16.45	**89.08±5.46**	88.13±2.68
Grass	77.05±4.09	84.59±4.39	81.40±5.78	84.43±1.59	76.89±2.42	87.51±2.91	41.72±14.29	49.06±12.47	89.46±7.80	**92.27±1.15**
Red roof	92.99±1.93	96.77±1.05	97.12±1.51	**98.24±0.39**	85.78±8.46	96.21±2.31	93.88±0.89	96.16±0.71	96.35±1.96	97.29±1.17
Gray roof	93.66±1.77	94.87±0.36	92.79±5.17	95.79±0.69	92.58±2.06	92.49±3.30	87.27±6.90	93.96±2.26	69.77±24.56	**96.93±0.74**
Plastic	67.15±7.02	78.10±7.65	84.54±6.21	87.84±3.09	87.54±5.82	89.71±5.60	58.03±6.67	71.89±10.27	**97.66±3.47**	97.64±2.56
Bare soil	66.58±2.27	74.04±2.36	68.13±8.97	71.04±1.41	68.88±4.94	70.01±4.56	26.95±12.44	62.26±3.06	80.44±6.01	**83.17±1.49**
Road	79.14±1.67	85.93±1.28	82.56±6.56	86.09±2.87	77.22±5.86	87.53±2.56	78.62±8.63	68.67±5.15	**94.20±2.54**	90.38±1.66
Bright object	81.55±2.09	84.44±3.69	95.00±2.41	97.10±0.51	88.92±3.94	89.60±2.43	90.07±2.34	83.47±2.76	**97.49±1.07**	82.17±4.55
Water	97.99±0.78	98.32±0.82	98.09±0.08	98.38±0.33	97.22±1.45	98.10±0.81	82.23±6.61	96.81±0.86	87.45±26.34	**98.73±0.56**
OA (%)	86.58±0.30	89.95±0.46	89.38±0.88	90.06±0.69	85.50±1.60	88.89±1.28	74.89±2.38	81.45±1.45	87.55±7.13	**93.89±0.58**
AA (%)	82.18±0.88	87.14±0.27	87.71±1.13	89.13±0.73	84.50±0.75	88.17±1.12	73.55±2.13	77.64±2.03	89.37±3.63	**92.89±0.68**
$\kappa \times 100$	84.36±0.35	88.28±0.52	87.63±1.02	88.43±0.80	83.18±1.81	87.08±1.48	71.23±2.60	78.48±1.69	85.68±7.71	**92.87±0.68**

TABLE IX

OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR THE WHUHH DATASET

	Patch-level						Image-level			
	DSNet [22]	SV^{2}Net [23]	DSFormer [27]	CSCANet [24]	3DSS-Mamba [18]	SliMamba [35]	MambaHSI [16]	MambaLG [26]	MambaMoE [27]	Proposed
Red roof	95.78±0.64	95.99±0.34	95.04±0.58	93.71±0.31	89.94±3.24	95.56±0.57	90.75±1.18	93.21±0.84	93.35±6.10	**96.46±0.32**
Road	94.29±1.66	94.48±1.31	92.52±1.18	95.38±1.26	90.95±3.95	92.58±0.82	86.87±8.84	88.24±0.49	**96.46±1.70**	91.36±2.56
Bare soil	82.58±0.99	85.29±0.68	85.08±1.58	88.22±2.23	86.29±1.28	86.07±2.08	83.60±0.98	83.47±4.16	88.66±2.40	**92.15±1.46**
Cotton	94.84±0.74	95.05±0.87	**96.05±0.54**	93.72±1.11	86.49±4.70	94.66±0.84	90.80±1.66	91.42±0.62	88.45±8.21	95.14±1.02
Cotton firewood	88.53±0.86	91.13±0.66	90.64±0.64	91.23±1.54	85.33±3.15	90.29±3.56	84.07±3.07	88.12±3.51	**94.27±4.11**	93.68±0.95
Rape	91.64±1.22	94.31±1.05	91.31±0.99	95.10±1.59	88.65±2.06	92.72±1.03	90.70±2.56	84.52±1.33	92.93±2.46	**97.09±0.60**
Chinese cabbage	63.49±3.32	68.20±3.63	78.78±3.88	**85.24±5.03**	61.46±7.50	77.56±3.01	72.85±7.15	60.63±2.07	70.90±10.00	82.88±3.04
Pakchoi	60.17±2.86	66.91±7.33	85.40±0.91	82.49±4.07	73.19±4.32	86.92±4.55	68.87±4.21	29.11±14.85	75.08±13.04	**96.73±0.88**
Cabbage	94.58±1.89	96.36±1.49	96.24±0.80	96.56±0.25	95.34±4.15	94.42±1.25	96.79±0.99	95.16±0.78	94.64±1.63	**97.59±0.90**
Tuber mustard	65.34±6.32	79.68±5.93	81.02±2.38	83.25±4.61	73.22±4.78	86.39±2.97	72.17±7.44	54.44±2.28	84.15±6.92	**91.46±1.98**
Brassica parachinensis	57.75±2.34	73.38±5.54	80.44±1.16	85.76±1.64	70.15±7.60	84.11±1.74	76.90±2.86	39.80±8.71	72.92±8.88	**88.50±2.54**
Brassica chinensis	65.25±7.39	74.98±3.30	68.04±2.63	79.19±2.10	62.18±6.81	72.92±2.41	63.66±5.61	61.42±3.62	82.67±4.58	**90.53±3.30**
Small Brassica chinensis	64.94±2.21	68.12±3.45	68.09±9.04	69.79±5.04	65.77±6.62	77.73±4.12	66.74±4.88	58.65±5.33	67.08±3.66	**85.98±0.99**
Lactuca sativa	74.99±3.19	87.15±2.22	88.08±3.73	93.08±2.66	90.42±2.07	92.66±1.33	90.59±6.01	61.95±0.88	84.96±5.25	**98.20±0.52**
Celtuce	98.36±0.66	98.36±0.79	**99.73±0.17**	98.27±0.38	98.29±1.02	98.91±1.04	97.32±1.43	94.43±2.94	99.68±0.29	98.82±0.85
Film covered lettuce	91.41±1.98	92.24±2.58	89.54±3.71	95.17±0.45	90.18±3.17	95.49±1.52	92.21±1.24	60.54±10.14	**97.96±0.71**	97.54±1.34
Romaine lettuce	88.61±3.50	94.65±2.13	96.97±1.49	**98.93±0.30**	85.75±12.75	94.81±1.80	90.80±4.07	84.68±4.51	94.03±1.90	92.91±5.03
Carrot	93.25±5.09	96.92±0.57	98.04±0.54	**99.03±0.35**	93.53±2.28	97.15±1.08	93.42±0.92	91.35±4.39	**97.62±1.10**	97.30±0.88
White radish	92.10±4.21	96.65±0.69	93.52±2.56	**96.70±1.19**	89.10±6.49	95.64±0.62	88.26±5.25	89.82±0.68	93.15±2.48	94.14±1.64
Garlic sprout	94.58±1.50	95.45±1.56	92.91±1.36	97.15±1.54	90.10±4.32	**97.90±0.81**	91.38±1.86	90.51±4.57	97.74±1.27	97.54±0.90
Broad bean	81.36±3.89	98.45±0.75	98.79±0.58	98.81±0.47	91.19±5.18	99.81±0.25	95.41±1.32	69.21±16.78	87.34±24.16	**99.95±0.06**
Tree	92.66±3.06	94.27±2.41	97.78±0.64	98.42±0.45	90.42±2.91	97.83±1.17	90.95±1.74	90.21±4.67	97.54±1.90	**99.24±0.49**
OA (%)	86.34±0.61	89.17±0.41	90.05±0.95	90.96±0.54	83.14±1.97	90.80±0.63	86.06±0.87	81.13±0.89	86.84±3.70	**93.73±0.41**
AA (%)	83.02±0.81	88.09±0.49	89.27±0.87	91.60±0.27	84.00±1.72	91.01±0.87	85.23±0.59	75.50±0.21	88.71±1.90	**94.33±0.65**
$\kappa \times 100$	82.93±0.76	86.47±0.49	87.53±1.17	88.72±0.66	79.34±2.23	88.51±0.78	82.74±1.01	76.62±1.05	83.85±4.23	**92.16±0.50**

content-aware routing of SSDM, and the physical unmixing representations of SNLMM.

1) Feature Activation Analysis of SSLT: As shown in Fig. 15, we compare the cosine similarity distribution between a reference patch (starred) and others. Compared to the baseline (left), the SSLT-processed map (right) demonstrates that Spatial-LA effectively amplifies deep correlations among semantically similar patches, yielding highly focused feature activations.

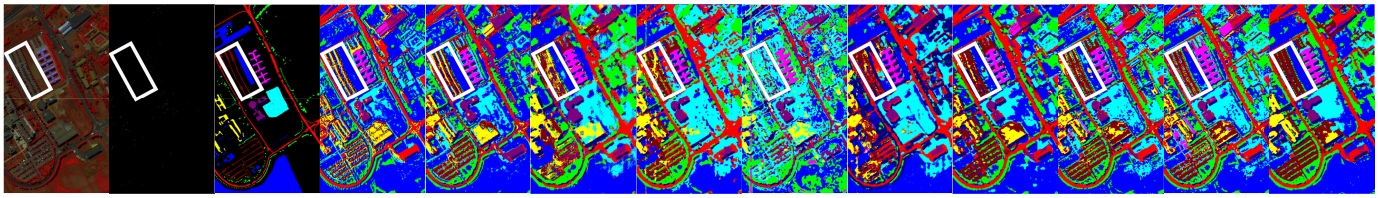
2) Adaptive Routing in SSDM: Fig. 16 illustrates the spatial deformable scanning in SSDM. Unlike the initial boundary-agnostic raster scan (left), the learned offsets (middle) assign

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TABLE X

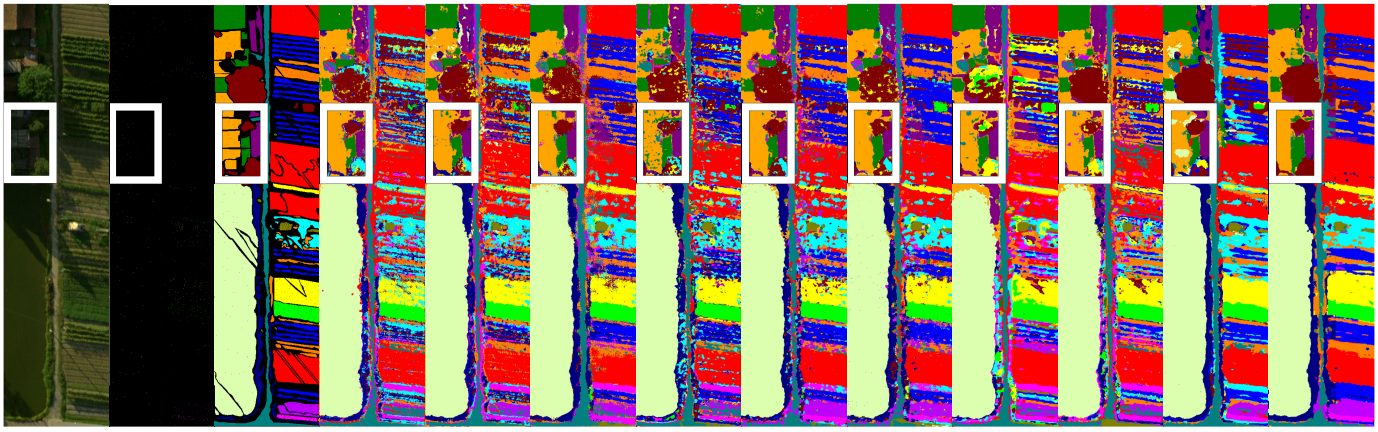
OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR THE AUGSBURG DATASET

	Patch-level						Image-level			
	DSNet [22]	SV^{2}Net [23]	DSFormer [27]	CSCANet [24]	3DSS-Mamba [18]	SliMamba [35]	MambaHSI [16]	MambaLG [26]	MambaMoE [27]	Proposed
Forest	94.35±0.78	97.23±0.24	91.81±0.27	91.74±2.82	86.38±7.61	92.02±1.89	95.80±1.10	97.12±0.35	83.75±6.35	**98.45±0.80**
Residential Area	94.48±0.99	94.92±1.03	**98.34±0.49**	96.66±0.87	89.33±6.84	98.06±0.62	97.00±0.84	96.44±0.17	82.04±3.42	96.95±1.37
Industrial Area	65.09±9.78	**72.64±9.69**	54.57±8.16	56.73±7.09	34.84±18.38	55.41±11.36	62.24±3.78	62.27±25.04	58.23±14.80	65.52±10.46
Low Plants	92.65±1.79	94.11±1.35	89.93±0.81	87.31±2.93	84.71±6.55	91.06±3.10	86.14±2.08	89.34±2.09	76.89±3.18	**95.41±0.91**
Allotment	51.85±5.38	49.98±9.77	67.49±2.04	59.39±3.22	39.50±6.89	65.91±2.75	34.38±7.95	47.99±8.18	65.05±6.12	**67.52±8.48**
Commercial Area	14.84±2.86	14.10±11.46	6.73±1.53	7.91±3.43	1.68±1.43	6.89±3.83	0.00±0.00	5.93±6.93	**25.15±12.65**	1.72±1.58
Water	28.19±9.03	30.13±14.98	41.98±5.66	46.05±2.88	11.72±6.67	41.70±9.14	0.38±0.48	42.60±1.85	43.78±4.09	**54.61±3.66**
OA (%)	89.12±0.82	90.67±0.60	88.93±0.41	87.52±1.39	80.85±4.28	89.27±0.53	87.00±0.99	89.15±1.84	77.33±2.04	**92.07±0.38**
AA (%)	63.06±1.92	64.73±2.69	64.41±1.28	63.68±1.61	49.74±3.31	64.44±0.49	53.71±1.60	63.10±4.02	62.13±1.63	**68.48±1.57**
$\kappa \times 100$	84.33±1.20	86.63±0.86	84.08±0.63	81.99±2.07	72.58±5.60	84.63±0.75	81.11±1.42	84.31±2.71	68.21±2.79	**88.59±0.53**



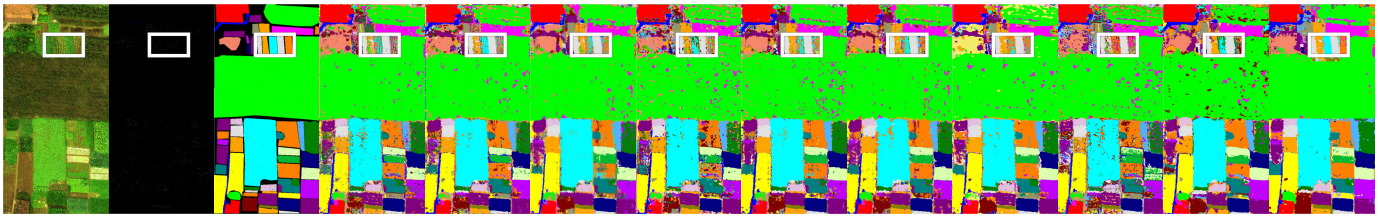
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m)

Fig. 10. False color, training and test set map of the PU dataset, and classification maps generated by different models (corresponding to the single experiment in Table VII). (a) False color. (b) Training set. (c) Test set. (d) DSNet. (e) SV^{2}Net. (f) DSFormer. (g) CSCANet. (h) 3DSS-Mamba. (i) SliMamba. (j) MambaHSI. (k) MambaLG. (l) MambaMoE. (m) Proposed.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m)

Fig. 11. False color, training and test set map of the WHUHC dataset, and classification maps generated by different models (corresponding to the single experiment in Table VIII). (a) False color. (b) Ground truth. (c) Test. (d) DSNet. (e) SV^{2}Net. (f) DSFormer. (g) CSCANet. (h) 3DSS-Mamba. (i) SliMamba. (j) MambaHSI. (k) MambaLG. (l) MambaMoE. (m) Proposed.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m)

Fig. 12. False color, training and test set map of the WHUHH dataset, and classification maps generated by different models (corresponding to the single experiment in Table IX). (a) False color. (b) Ground truth. (c) Test. (d) DSNet. (e) SV^{2}Net. (f) DSFormer. (g) CSCANet. (h) 3DSS-Mamba. (i) SliMamba. (j) MambaHSI. (k) MambaLG. (l) MambaMoE. (m) Proposed.

adjacent scanning priorities to spatially separated but semantically similar regions (red/yellow boxes) while maintaining the coherence of continuous land covers (blue box). The resulting dynamic trajectory (right) breaks geometric constraints to establish content-aware routing, effectively capturing long-range semantic dependencies.

3) Physical Unmixing Representations of SNLMM: As shown in Fig. 17, SNLMM generates interpretable fractional abundance maps and physically meaningful endmembers. The abundance maps are spatially consistent with the classification ground truth, while the extracted spectral curves reflect material-related signatures. By integrating classification with

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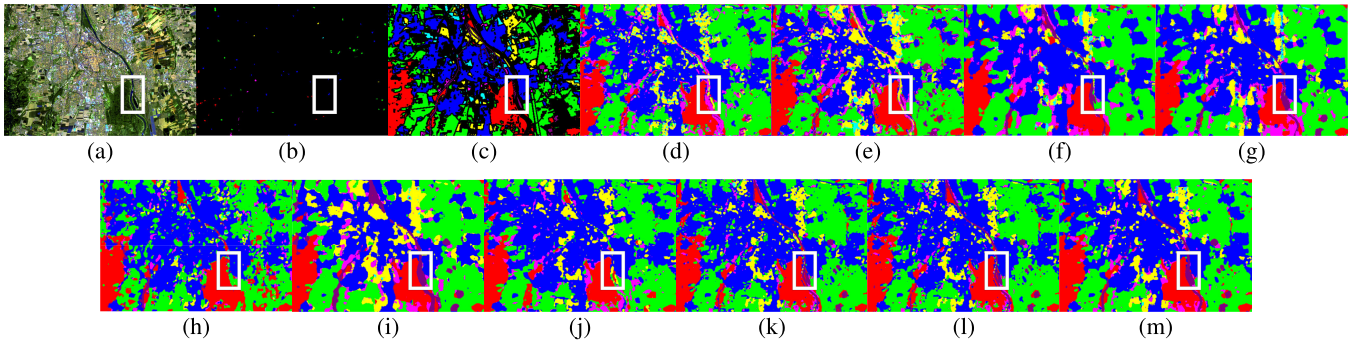


Fig. 13. False color, training and test set map of the Augsburg dataset, and classification maps generated by different models (corresponding to the single experiment in Table X). (a) False color. (b) Ground truth. (c) Test. (d) DSNet. (e) SV^{2}Net. (f) DSFormer. (g) CSCANet. (h) 3DSS-Mamba. (i) SliMamba. (j) MambaHSI. (k) MambaLG. (l) MambaMoE. (m) Proposed.

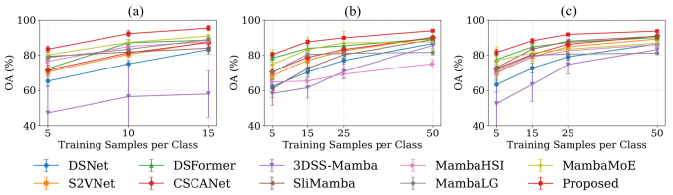


Fig. 14. Comparison of classification performance with limited training samples. (a) PU. (b) WHUHC. (c) WHUHH.

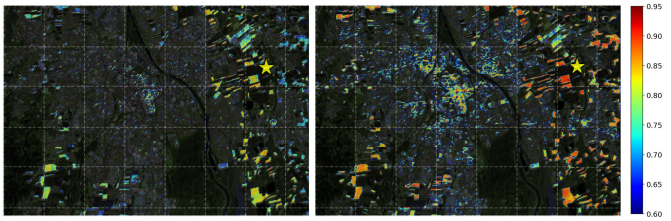


Fig. 15. Visualization of feature similarity before and after applying the SSLT block on the Augsburg dataset.

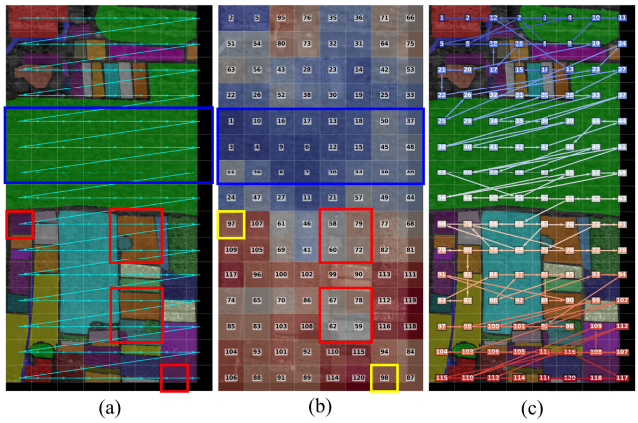


Fig. 16. Visualization of the spatial deformable scanning mechanism within the SSDM block on the WHUHC dataset. (a) Initial raster scan. (b) Learned spatial offsets. (c) Final spatial deformable scanning trajectory.

nonlinear unmixing, SNLMM provides a physically constrained subpixel representation beyond purely task-specific features. The interpretability of SNLMM comes from both its outputs and its physical constraint. The estimated abundance maps make class-wise subpixel material distributions observable, and the extracted endmember curves provide direct spectral meanings for the learned representation. In addition,

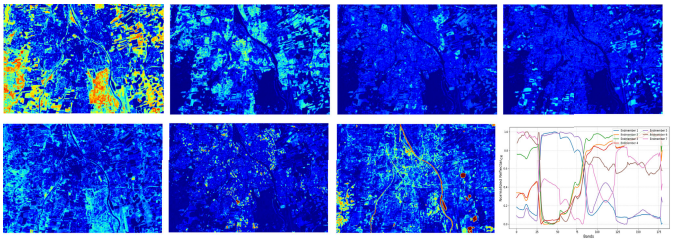


Fig. 17. Estimated fractional abundance maps and extracted endmember spectral signatures on the Augsburg dataset.

TABLE XI

MODEL COMPLEXITY AND EFFICIENCY COMPARISON

	Patch-level Models						Image-level Models			
	DSNet	SV^{2}Net	DSFormer	CSCANet	3DSS-Mamba	SliMamba	MambaHSI	MambaLG	MambaMoE	Proposed
Parameters (K)	154.24	401.02	268.96	263.57	**10.31**	110.62	16.14	132.33	818.32	66.21
FLOPs (G)	2.65	1.14	1.67	250.33	4.38	3.96	0.42	2.03	1.77	**0.374**
Time$_{tr}$	111.23	203.84	59.68	157.97	59.68	67.74	119.93	164.23	52.26	**51.32**
Time$_{te}$	21.5	65.15	6.17	7.37	6.79	3.45	0.23	2.26	0.28	**0.21**

the nonlinear reconstruction loss constrains the abundance-endmember combination to reconstruct the raw HSI, which reduces the risk of learning arbitrary latent features inconsistent with the observed spectra. These results show that SNLMM serves as a physically grounded subpixel representation module rather than only a discriminative auxiliary branch.

V. CONCLUSION AND FUTURE RESEARCH

In this article, we proposed I2SNet, a hierarchical coarse-to-fine network designed for accurate and efficient HSIC. By systematically extracting representations across patch, pixel, and subpixel scales, the proposed architecture alleviates the restricted receptive fields of patch-based methods and the coarse feature processing of traditional image-level networks. The synergistic integration of the SSLT and the SSDM effectively captures global spatial–spectral dependencies. Furthermore, the SNLMM explicitly decodes the physical mixing mechanisms, mitigating subpixel blindness and yielding discriminative features. Comprehensive experiments on four benchmark datasets confirm that I2SNet delivers strong classification performance while maintaining a compact computational footprint. Nevertheless, the classification accuracy for categories with extremely limited training samples remains challenging, and future work will further investigate minority-class enhancement and uncertainty-aware learning strategies

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to improve the robustness of I2SNet in such cases. Future research will focus on simplifying the network architecture to further reduce model parameters, as well as extending the framework to multimodal remote sensing data fusion for more challenging observation scenarios.

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